

Recursive Fractal Cosmology: A Triadic Ontology of Existence

Allan Edward

June 5, 2026

Contents

1	Introduction: the problem this rebuild solves	1
1.1	What changed from the first three preprints	2
1.2	Claim-status vocabulary	2
2	The triadic scientific ontology	3
2.1	CIF: modal source space	3
2.2	QV: compression, damping, and selection	3
2.3	RFL: stabilized recursive output	3
2.4	The first action	4
3	Recursive kernel and modal basis	4
3.1	Why Feigenbaum scaling appears	5
3.2	Entropy as recursive memory	5
4	Deterministic closure spine	5
4.1	Module G: frozen triadic packet	6
4.2	Module R V3: source-coupled closure	6
4.3	Module N V2: dimensional projection bridge	7
4.4	Modules S and T: SI bridge and dimensionless coupling map	7
5	Upgraded recursive module architecture	7
5.1	Module 1: recursive inflation and reheating	8
5.2	Module 2: QCD and electroweak thermal transitions	9
5.3	Module 3: domain walls and defect/kink geometry	9
5.4	Module 4: primordial black-hole and overdensity channels	9
5.5	Module 5: neutrino background and effective degrees of freedom	10
5.6	Module 6: CP phase, EDM windows, and baryogenesis	10
5.7	Module 7: CMB visibility, recombination, and transparency	11
5.8	Module 8: particle masses and Standard-Model-facing structure	11
5.9	Module 9: PMNS/CKM mixing and frozen particle-sector retest	12
5.10	Module 10: decoherence, observer branch separation, and neural signatures	12
6	New boundary lanes beyond the module architecture	12
6.1	Y3: a model repair done correctly	13
6.2	U/U2-U5 and P2: separating scaffold from constants	13
6.3	V2: precision cosmology boundary	13
6.4	W2: BBN and the lithium wall	14
6.5	X2: CP, EDM, and baryogenesis boundary	14
6.6	GW2: ringdown and echo window	14

6.7	Z2: observer and neural analysis	14
6.8	E2: early-universe relic and transition registry	14
7	Quantitative spine and internal screen summary	15
8	Public-data and public-code handoff	16
8.1	Cosmology handoff	16
8.2	BBN handoff	16
8.3	Particle and constants handoff	16
8.4	CP/EDM/baryogenesis handoff	16
8.5	Gravitational-wave and neural handoffs	16
9	Figures, tables, and page architecture	16
10	Falsifiability and blocked claims	17
11	Discussion	17
12	The collapse-rebirth cycle across the module system	18
13	Conclusion	19
A	Full boundary module ledger	21
B	Symbol glossary	21
C	External-run registry	22
D	Module-specific boundary dossier	22
D.1	G: deterministic triadic closure	22
D.2	R V3: source-coupled closure audit	23
D.3	N V2: dimension-aware projection bridge	23
D.4	S/T: unit and dimensionless coupling interface	24
D.5	Y3: frozen particle-sector retest	25
D.6	U5: Standard-Model-facing scaffold	25
D.7	P2: constants and parameter-table boundary	26
D.8	V2: precision cosmology external-likelihood lane	26
D.9	W2: BBN and Li7 boundary	27
D.10	X2: CP/EDM/baryogenesis source-window	27
D.11	GW2: gravitational-wave waveform boundary	28
D.12	Z2: observer, neural, and EEG boundary	28
D.13	E2: early-universe relic and transition registry	29
D.14	QG/QG2/J2: quantum-geometry and spin-foam audit	29

D.15 LEG2: legacy reconciliation	30
D.16 TK2: manuscript assembly controller	30
E Scientific-language conversion table	32
F Detailed external-run protocols	32
F.1 V2 protocol: cosmology	32
F.2 W2 protocol: BBN	33
F.3 X2 protocol: EDM and baryogenesis	33
F.4 Y3/U5/P2 protocol: particle sector and constants	33
F.5 GW2 protocol: waveform analysis	33
F.6 Z2 protocol: neural and observer signatures	33
G Visual and table production plan	33
H Expanded module dossier	34
H.1 Triad foundation: CIF/QV/RFL definitions	34
H.2 Kernel: recursive modal operator	35
H.3 G: deterministic closure	35
H.4 R V3: closure audit	36
H.5 N V2: dimensional projection	37
H.6 S/T: unit bridge	38
H.7 Y3: particle retest	38
H.8 U5/P2: SM scaffold and constants boundary	39
H.9 V2: precision cosmology	40
H.10 W2: BBN	41
H.11 X2: CP/EDM/baryogenesis	41
H.12 GW2: waveforms	42
H.13 Z2: neural/observer	43
H.14 E2: early universe	44
H.15 QG: quantum geometry	44
H.16 LEG2/TK2: legacy and assembly	45
I Detailed claim-boundary registry	47
J Reproducibility	48
J.1 Module G - Deterministic triadic closure	48
J.2 Module R V3 - Source-coupled triad closure audit	49
J.3 Module N V2 - Dimensional projection bridge	49
J.4 Modules S/T - SI bridge and dimensionless coupling map	50
J.5 QG/QG2/J2 - Spin-foam and quantum-geometry suite	51

J.6	Y3 - Frozen particle-sector retest	52
J.7	U/U2-U5 - Standard-Model-facing closure ledger	53
J.8	P2 - Constants and parameter-table boundary	54
J.9	V2 - Precision cosmology boundary	56
J.10	E2 - Early-universe relic and transition boundary	58
J.11	W2 - BBN and lithium-7 boundary	60
J.12	X2 - CP, EDM, and baryogenesis boundary	62
J.13	GW2 - Gravitational-wave and ringdown boundary	64
J.14	Z2 - Observer, neural, and EEG boundary	66
J.15	LEG2 - Legacy reconciliation ledger	68
J.16	Combined downstream physical-projection screen	70
J.17	Dimensionless observable validation layer	73
J.18	TK2 - Canonical preprint assembly ledger	73

References	76
-------------------	-----------

List of Figures

1	Scientific interpretation of the CIF/QV/RFL triad. The terms are used as modeling roles: source space, compression/selection operation, and stabilized recursive output.	1
2	Kernel architecture. The same damping/scaling skeleton is combined with different modal bases for cosmology, particle sectors, thermal transitions, gravitational waves, and neural analysis.	4
3	Dependency order for this preprint. The dependency order should never allow a downstream module to tune the source packet.	5
4	Module R V3 groups source power into QV, CIF, RFL, and dark-kernel bridge components. The closure audit is QV-dominant, with the dark-kernel bridge carrying most of the remaining power.	7
5	Triad-to-module dependency map. The numerical values are qualitative design scores for how strongly each module depends on CIF modal content, QV damping, RFL output, internal pass status, and external-run readiness.	8
6	Y3 repairs a specific exploratory weakness under a frozen retest. The figure should be read as an internal repair, not as a full particle-physics derivation.	13
7	Downstream boundary lanes. A lane can be valuable even when it is not externally validated, if it specifies the public data, code, outputs, and blocked claims.	15
8	Recommended page architecture from the page-map analysis. The target is roughly 72 pages, with the main body carrying the triad, kernel, deterministic closure, module architecture, and boundaries before references and appendices.	17
9	Collapse-rebirth cycle generated by the module system. The Big Implosion begins the cycle as QV compression of CIF potential. The kernel distributes that compression into domain-specific modal bases. The downstream modules then describe how expansion, transitions, asymmetry, visibility, particles, decoherence, and memory become stabilized RFL outputs. The return arrow is a memory return, not a completed claim of cyclic cosmology.	18

List of Tables

1	Claim-status table used throughout this preprint.	3
2	Frozen Module G packet.	6
3	Core internal quantities and their allowed interpretation.	15

Abstract

Abstract

Recursive Fractal Cosmology (RFC) begins from a triadic claim: emergence can be modeled only when source, operation, and stabilized output are kept distinct. The Cosmic Infinite Field (CIF) is treated as modal source space; the Quantum Vacuum (QV) is the compression, damping, and selection operation; and the Recursive Fractal Lattice (RFL) is the memory-bearing structure through which geometry, fields, particles, observers, and cosmological history become organized. In its shortest form, the first action is $QV(CIF) \rightarrow RFL$. The Big Implosion is the first cosmological expression of that action: recursive compression of modal possibility into structured output.

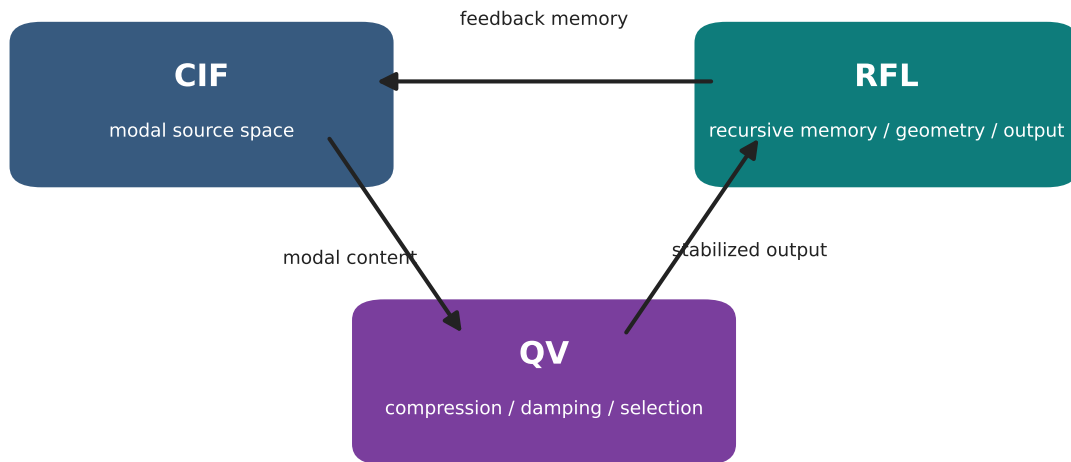
Earlier RFC preprints developed this story in stages: symbolic cosmology, kernel/module architecture, mathematical codex, and deterministic closure. This paper consolidates that sequence into a scientific rebuild. It freezes a no-target triadic packet before comparison, carries it through source-coupled closure, dimensional projection, module equations, boundary lanes, collapse-rebirth logic, and reproducibility appendices, and marks each result by claim status. The aim is not to claim external validation for every observable; it is to provide a no-retune framework whose next public tests are explicit.

Prior RFC preprints

DOI sequence: 10.5281/zenodo.15205532; 10.5281/zenodo.15234638;
10.5281/zenodo.15250393; 10.5281/zenodo.20473033.

Central thesis

Dependency order: triad $\rightarrow$ kernel $\rightarrow$ deterministic packet $\rightarrow$ closure audit $\rightarrow$ dimensional bridge $\rightarrow$ external-run lanes. External data enter after the frozen packet, not before it.



Canonical action: $QV(CIF) \rightarrow RFL$

Figure 1: Scientific interpretation of the CIF/QV/RFL triad. The terms are used as modeling roles: source space, compression/selection operation, and stabilized recursive output.

1 Introduction: the problem this rebuild solves

The earlier RFC sources developed a large theory space. They proposed a triadic origin, a recursive cosmology, a fractal measure, a Big Implosion mechanism, CP-asymmetric local branches, dark-sector interpretations, symbolic field carriers, spin-foam weighting, observer fields, and a family of domain modules. That breadth produced explanatory power. It also created a problem: if every module is

allowed to speak with the same strength, the paper can sound more certain than the calculations justify.

This preprint is written to solve that problem. It keeps the explanatory arc of the early preprints, especially the clarity of the first preprint and the module flow of the second preprint. It also keeps the main repair introduced by the fourth preprint: the active model must follow a no-retune rule. A packet derived internally from the triad and recursive closure must be frozen before downstream comparison. If a module later needs public data, external solvers, likelihoods, or laboratory constants, that requirement is stated as a boundary rather than disguised as a completed result.

This document therefore has two jobs. First, it gives a pure-scientific reconstruction of the triad. Second, it maps the rebuilt module system into a claim ledger. A claim ledger is not a rhetorical hedge. It is a way to make the paper falsifiable: each claim is tied to a status, and each stronger claim names the work still needed.

1.1 What changed from the first three preprints

The first preprint supplied the strongest story: QV compresses CIF into RFL, and the Big Implosion becomes the first cosmological expression of that compression. The second preprint supplied the strongest architecture: one recursive kernel, many modal bases, and a sequence of physical modules. The third preprint supplied mathematical breadth: categorical, spin-foam, symbolic, and integrable structures. The fourth preprint supplied discipline: deterministic closure, frozen packet, no MCMC/NUTS upstream fitting, dimensional projection separation, and explicit boundaries.

This preprint must combine these rather than choose among them. The opening must not begin with a table of numbers. The reader needs to understand why three roles are necessary. After that, the theory can safely introduce equations, modules, closure scores, and public-data requirements. The goal is a paper that is both legible and harder to overclaim.

1.2 Claim-status vocabulary

The most important editorial move is to stop using a single word, “validated,” for different things. This draft uses five categories:

1. **Internal derivation:** a quantity follows from the frozen packet and algebraic/procedural rules inside RFC.
2. **Internal closure:** a source-coupled audit or identity check closes under the no-retune rules.
3. **Proxy screen:** an internally projected or compressed number lies in a meaningful window but is not a public likelihood result.
4. **External-run ready:** the module defines the public data, code, outputs, and decision criteria required for validation.
5. **Blocked claim:** a stronger claim is explicitly disallowed until a specified external calculation is performed.

Table 1: Claim-status table used throughout this preprint.

Status	It permits	It does not permit
Internal derivation	Reporting frozen packet values and internal identities	Claiming public constants or external data fits
Internal closure	Reporting no-retune closure audits	Claiming empirical confirmation
Proxy screen	Reporting bounded internal windows	Treating proxies as measurements
External-run ready	Specifying datasets, solvers, and outputs	Claiming the run has already passed
Blocked claim	Explaining what is not yet justified	Hiding unresolved failures

2 The triadic scientific ontology

In this preprint, “ontology” means a minimal set of modeling roles, not a completed metaphysical proof. RFC begins by asserting that emergence needs three functions. A source function supplies possible structure. A selection function compresses and actualizes part of that structure. A memory function stabilizes selected structure into a persistent output. RFC names these three functions CIF, QV, and RFL.

2.1 CIF: modal source space

The Cosmic Infinite Field is used here as the space of modal possibilities. It is not treated as a directly measured field in the same sense as a Standard Model field. It functions as the domain from which recursive modes are drawn. In early RFC language CIF carried “infinite possibility.” In this preprint the same idea is rewritten as a source-space assumption: before a module can select, damp, or stabilize a mode, there must be a mathematical space of possible modes.

This is why CIF appears most strongly in the modal function $f_j(t)$, in phase choices, in eigenfield language, in particle-sector mixing, in CP-phase windows, and in the branching language of observer modules. It supplies available alternatives. It does not by itself decide which alternative becomes physical.

2.2 QV: compression, damping, and selection

The Quantum Vacuum is used as the operation that selects, damps, and compresses modal content. The term should not be read as a claim that standard quantum field theory already contains the whole RFC mechanism. It marks the role played by local fluctuation, decay, entropy-gradient formation, and branch selection. The QV role is therefore operational: it turns modal space into directed recursion.

This role explains why the QV side of RFC carries exponential factors, entropy gradients, collapse/re-birth variables, CP residuals, and source-coupled residual dominance. In Module R V3 the QV core becomes the dominant closure driver. That is not an aesthetic preference. It is a result of the closure audit: entropy flow and entropy recoil dominate the source-grouped residual structure.

2.3 RFL: stabilized recursive output

The Recursive Fractal Lattice is the persistent output of the triadic action. RFL is where geometry, structure, memory, and observability appear. Earlier RFC language called it the emergent child of QV and CIF. In scientific language it is the structured state space produced when modal source content is compressed into stabilized recursion.

RFL is not passive. It feeds back as memory. Once a recursive output stabilizes, it changes the later selection environment. This is why the RFL appears in spin-foam weights, geometry coherence, observer memory, rebirth memory, particle-sector gates, and external observables.

2.4 The first action

The first action is written as

$$\text{QV}(\text{CIF}) \longrightarrow \text{RFL}. \quad (1)$$

This equation should not be inflated. It is a dependency statement: modal content is acted on by a selection/compression operation, and the result is stabilized recursive output. It is also the reason the paper must not begin with Module G. Module G is downstream. The triad is the primitive grammar.

3 Recursive kernel and modal basis

The mathematical core must be compact enough to survive scrutiny. RFC's module architecture is built around a universal recursive kernel with domain-specific modal functions:

$$K_f(t) = \sum_{j=1}^N \delta^{-j} f_j(t) e^{-\alpha j t}. \quad (2)$$

Here δ is the Feigenbaum scaling constant used as the recursive-bifurcation scale, α is a damping rate derived in the frozen packet, j indexes recursive depth, and $f_j(t)$ carries the domain-specific modal basis. The key architectural claim is not that every domain has the same observable equation. The claim is that each domain inherits the same recursion skeleton while using a different modal basis.

Universal recursive kernel with domain-specific modal bases

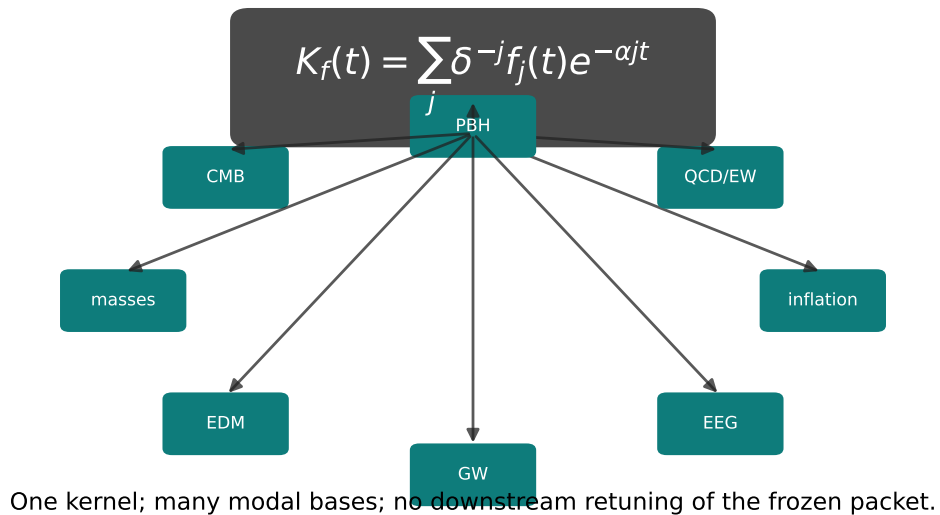


Figure 2: Kernel architecture. The same damping/scaling skeleton is combined with different modal bases for cosmology, particle sectors, thermal transitions, gravitational waves, and neural analysis.

The triad-to-kernel mapping is direct. CIF contributes the modal basis f_j . QV contributes the damping and selection envelope $e^{-\alpha j t}$ and the recursive weighting. RFL is the realized output after

summation and stabilization. This mapping is important because it prevents the triad from becoming ornamental. Each term has a job in the computation.

3.1 Why Feigenbaum scaling appears

Feigenbaum scaling is used because RFC treats emergence as recursively bifurcating structure. The use of δ is not presented as proof that all physical systems literally follow the logistic map. It is used as a symbolic-universal scaling anchor for recursive bifurcation, consistent with the broader universality literature on period-doubling and nonlinear dynamics [1, 2, 3].

The strong version of the claim is blocked: RFC does not yet demonstrate that every physical module reduces to a rigorously proven Feigenbaum universality class. The safe version is narrower: RFC uses δ as a deterministic internal scaling primitive, then audits whether the resulting packet remains coherent across module projections.

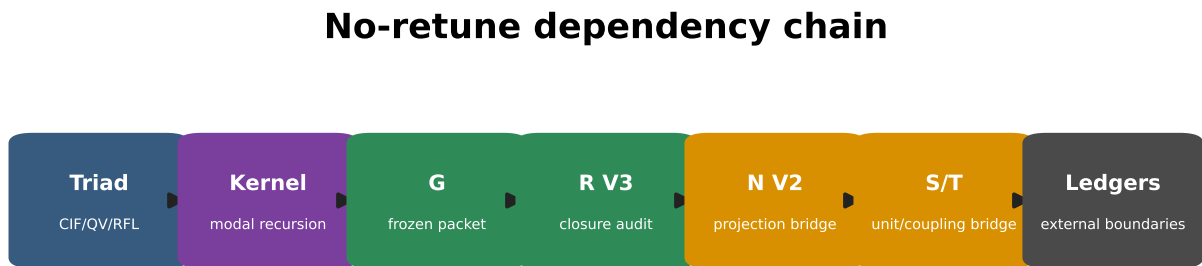
3.2 Entropy as recursive memory

In RFC, entropy is not treated only as disorder. It is used as a memory-bearing quantity that records recursive bifurcation history. This interpretation should be handled carefully. The conventional thermodynamic meaning remains intact where public physics requires it. The RFC-specific claim is that the recursive kernel adds a memory interpretation: entropy gradients track how modal possibilities have been selected into stabilized outputs.

This interpretation matters for time. The arrow of time is not appended after the fact. It appears as the directed record of recursive compression. That is why QV damping, entropy flow, collapse/rebirth, and RFL memory sit in the same dependency chain.

4 Deterministic closure spine

The fourth preprint's greatest improvement was methodological. It replaced the older habit of fitting or selecting parameters through downstream targets with a deterministic source packet. The boundary ledgers preserve and sharpen this rule.



External data enter after the frozen internal packet, not before it.

Figure 3: Dependency order for this preprint. The dependency order should never allow a downstream module to tune the source packet.

4.1 Module G: frozen triadic packet

The current frozen packet is:

$$\delta = 4.6692, \quad (3)$$

$$L = 60, \quad (4)$$

$$\alpha = \frac{\log \delta}{L} = 0.0256831, \quad (5)$$

$$k = 2, \quad (6)$$

$$\nu = 2\delta^{-4} = 0.00420784, \quad (7)$$

$$\epsilon = \alpha\nu = 0.000108071, \quad (8)$$

$$\lambda_N = 0.489442, \quad (9)$$

$$n_{\text{closure}} = 18, \quad (10)$$

$$n_{\text{full}} = 40. \quad (11)$$

No empirical targets are used to choose these values in the active rebuild. That statement is the spine of the preprint. The paper can later compare to public data, but it must not move the packet after seeing the comparison.

Table 2: Frozen Module G packet.

Quantity	Value	Role
δ	4.6692	Recursive bifurcation scale
L	60	Cycle length
α	0.0256831	Damping rate, $\log(\delta)/L$
k	2	Phase-depth input
ν	0.00420784	$2\delta^{-4}$
ϵ	0.000108071	$\alpha\nu$
λ_N	0.489442	Normalized internal closure quantity
n_{closure}	18	Closure depth
n_{full}	40	Canonical full depth
Empirical targets	False	No upstream fitting

Blocked move

Do not take the frozen packet and then silently adjust it to improve CMB, particle, BBN, EDM, or EEG behavior. Any such adjustment would return the framework to the older fit-first logic that this preprint is designed to avoid.

4.2 Module R V3: source-coupled closure

Module R audits how the frozen packet behaves when projected through source channels. The important improvement is conceptual: the residual is not treated as a bare number to minimize. It is decomposed into source groups. That source coupling lets the theory ask which part of the triad is doing the closure work.

The V3 interpretation is QV-dominant. The grouped source powers are approximately:

- QV entropy flow plus recoil: 0.794508,
- dark-kernel bridge: 0.19036,
- RFL observer/rebirth memory: 0.010047,
- CIF phase plus CP: 0.00508468.

The closure summary carries residual improvement $V2 = 0.418175$, source-coupled residual score $V2 = 0.576064$, memory transfer score = 0.146153, best lag = -4, best lag correlation = 0.873361, CP residual $\approx 3.59 \times 10^{-10}$, and Module R score $V2 = 0.726401$.

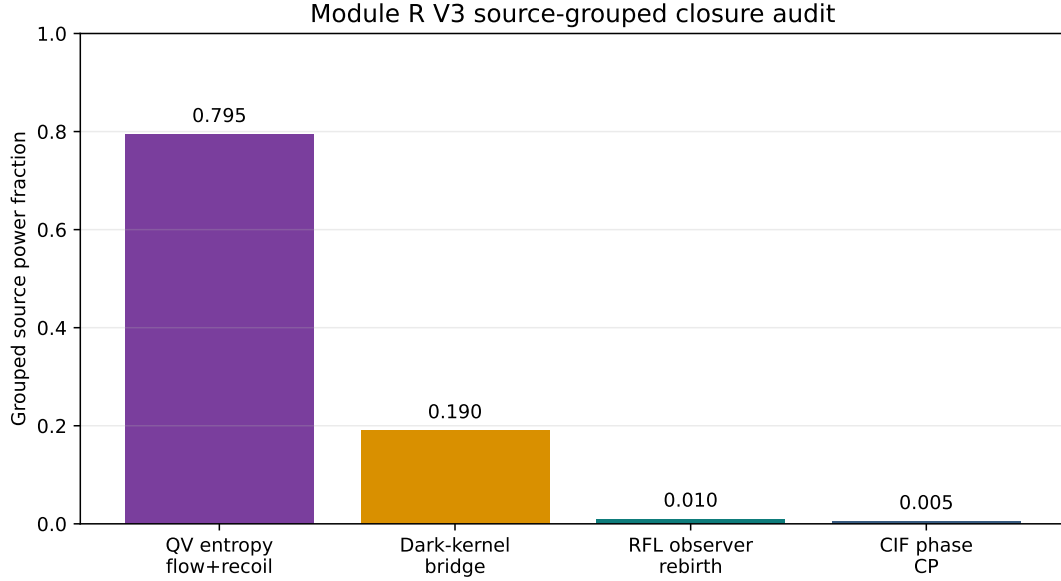


Figure 4: Module R V3 groups source power into QV, CIF, RFL, and dark-kernel bridge components. The closure audit is QV-dominant, with the dark-kernel bridge carrying most of the remaining power.

4.3 Module N V2: dimensional projection bridge

Module N V2 corrects a major internal risk: symbolic constants, internal projected units, and public SI constants cannot be conflated. The current bridge uses a dimension-aware projection rather than multiplying every symbolic value by the same factor.

The canonical bridge carries an energy factor of 0.28443 and a time factor of 1.54099. In internal RFC units the projected values include symbolic mass projected ≈ 0.0465 , mean energy projected ≈ 0.00438 , inverse-energy alpha projection ≈ 228 , mass-squared alphaG projection ≈ 0.00216 , and projected cycle time ≈ 92.46 . The identity residuals are near zero. The result is an internal dimensional consistency pass, not a derivation of public constants.

4.4 Modules S and T: SI bridge and dimensionless coupling map

S and T make the external interface explicit. S handles a one-anchor SI bridge; T handles dimensionless coupling comparison. This split is essential. Physical validation must be done in dimensionless ratios or through a transparent calibration anchor. Without that separation, the paper could accidentally present internal units as laboratory values.

5 Upgraded recursive module architecture

The second preprint remains the best guide for flow. It made the reader feel how one kernel can move through many physical domains. The boundary-ledger work improves it by adding dependency discipline, public-data boundaries, and no-retune claim labels. The next sections retain the second preprint sequence but rebuild its claims.

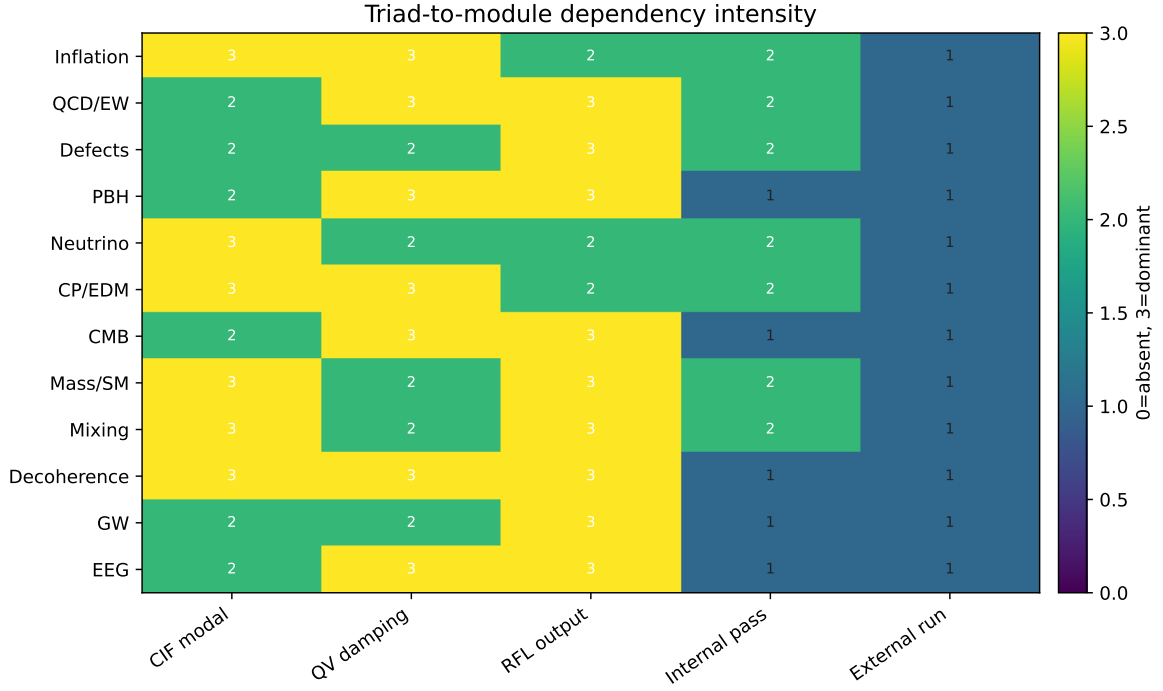


Figure 5: Triad-to-module dependency map. The numerical values are qualitative design scores for how strongly each module depends on CIF modal content, QV damping, RFL output, internal pass status, and external-run readiness.

5.1 Module 1: recursive inflation and reheating

Purpose. This module tests whether the triadic kernel can encode a primordial phase in which expansion, memory, and damping are not separate assumptions but coupled projections of the same recursion.

Role in the second preprint. The second preprint gave this module its best narrative role: inflation was presented as a recursive bifurcation from the same kernel later used for transitions, particle sectors, and decoherence. The value of that presentation is flow. The reader sees the first cosmological module as the first large-scale expression of the triad rather than as a detached calculation.

Upgrade path from the boundary ledgers. The E2 ledger upgrades the module by converting the old inflation language into a thermal-history boundary. It retains inflation-memory and reheating-entropy proxies but requires a public-code comparison before any external claim can be made. The upgraded module should therefore report internal proxies, then state exactly what is required: an inflationary background solver, reheating parameter scan, perturbation calculation, and comparison to CMB likelihoods.

Current status. External-run-ready, not solved inflation. Key internal quantities to carry are the cycle length 60, scalar-tilt proxy 0.964, tensor-ratio proxy 0.0017, running proxy -0.00056, inflation memory 0.717, and reheating entropy 0.146. These are internal or proxy quantities. They are not Planck likelihood outputs.

Boundary language. Says: RFC defines a no-retune inflation/reheating handoff. Doesn't say: RFC has replaced inflation or solved reheating.

5.2 Module 2: QCD and electroweak thermal transitions

Purpose. This module asks whether recursive damping and modal selection can parameterize finite-temperature transitions without changing the frozen packet.

Role in the second preprint. The second preprint placed QCD and electroweak transitions directly after inflation, which is good ordering. The reader moves from early expansion into thermal reorganization. The triad split is natural: CIF supplies phase-space alternatives, QV supplies damping and selection, RFL records the transition geometry.

Upgrade path from the boundary ledgers. E2 improves this module by requiring the transition story to be written as a handoff to lattice-QCD and electroweak finite-temperature literature rather than as a completed proof. The boundary language preserves the explanatory role while preventing unsupported claims about transition order, sphaleron freeze-out, or relic production.

Current status. Thermal-history boundary. Carry QCD crossover proxy 0.925, electroweak thermal-strength proxy 0.046, CP source proxy 0.545, washout proxy 0.533, and the no-retune packet from Module G.

Boundary language. Says: the module organizes finite-temperature transition targets. Doesn't say: RFC derives the QCD or electroweak transition from first principles.

5.3 Module 3: domain walls and defect/kink geometry

Purpose. This module studies localized recursive memory in field space: kink, wall, and defect-like structures as RFL-stabilized residues of modal selection.

Role in the second preprint. The second preprint made domain walls visually and conceptually useful because they show how an abstract recursive kernel can become a spatially structured object. This is one of the best bridges between explanation and physics.

Upgrade path from the boundary ledgers. The boundary ledgers move defects into the E2 and boundary framework. The upgraded version should keep the soliton/kink intuition but explicitly separate toy PDE behavior from cosmological defect constraints.

Current status. Internal structure plus external constraint target. Use the kernel form, defect/coherence language, and relic-stability proxy 0.925. For external work, cite topological-defect reviews and specify observational constraints on domain walls rather than treating RFC walls as automatically physical defects.

Boundary language. Says: defect-like modes are one modal basis of the kernel. Doesn't say: the model predicts observable domain walls without a constraint calculation.

5.4 Module 4: primordial black-hole and overdensity channels

Purpose. This module asks how recursive compression could be represented as early overdensity, collapse, or PBH-like seeding without changing the source packet.

Role in the second preprint. The second preprint linked PBH fields to recursive density kernels. That was valuable because it gave a concrete object to the collapse side of the triad.

Upgrade path from the boundary ledgers. E2 upgrades PBH language by placing it in a relic/transition boundary lane. The new paper should use PBHs as an external-run target: compute a power spectrum or collapse threshold, then compare against PBH abundance constraints. It should not infer PBH viability from internal density proxies alone.

Current status. External-relic calculation required. Carry PBH fraction proxy $3.4e-5$, relic stability proxy 0.925, and the dark-kernel bridge fraction 0.19036 from Module R V3. These values motivate a target, not a detection.

Boundary language. Says: RFC supplies a PBH/relic handoff. Doesn't say: RFC explains dark matter through PBHs unless external abundance constraints pass.

5.5 Module 5: neutrino background and effective degrees of freedom

Purpose. This module examines whether neutrino decoupling, mixing, and effective relativistic degrees of freedom can be described as modal-basis transitions under the same kernel.

Role in the second preprint. The second preprint connected neutrino freezing and PMNS structure to recursive eigenfields. That made the neutrino sector a natural bridge between early-universe thermodynamics and particle-sector modules.

Upgrade path from the boundary ledgers. The boundary ledgers preserve neutrino content in E2, Y3, and U5. They add the rule that neutrino quantities must be compared to NuFIT/PDG/BBN/CMB references and must not be selected by downstream target fitting.

Current status. Internal packet plus external comparison required. Use Neff placeholder 3.046, eta10 proxy 6.068 where relevant, and Y3 particle-sector retest logic for mixing. External validation requires NuFIT and cosmological constraints.

Boundary language. Says: the module defines a no-retune neutrino-sector comparison. Doesn't say: RFC derives all neutrino masses and mixings.

5.6 Module 6: CP phase, EDM windows, and baryogenesis

Purpose. This module is the main asymmetry lane. It tests whether CP-phase structure can be carried by the triadic kernel into baryon-asymmetry and EDM observables.

Role in the second preprint. The first and second preprints gave CP violation strong explanatory power: local asymmetry can be produced while global branch symmetry is preserved. The second preprint made this a module rather than only a story.

Upgrade path from the boundary ledgers. X2 is a major upgrade. It converts old CP/EDM/baryogenesis language into a seven-step source-window handoff. It preserves CP-phase/Jarlskog/theta proxies while requiring EFT operators, Wilson coefficients, physical EDM conversion, transport equations, and public bounds before stronger claims are allowed.

Current status. Source-window and external-boundary pass. Carry J proxy about 0.32e-4, eta10 proxy 6.068, CP residual about 3.59e-10, electron/neutron/atomic EDM dimensionless proxies at n=18 and n=40, memory/source/washout proxies, and 6/6 final X2 pass after repaired A2 path.

Boundary language. Says: RFC supplies a bounded CP source-window protocol. Doesn't say: RFC solves baryogenesis or predicts physical EDMs yet.

5.7 Module 7: CMB visibility, recombination, and transparency

Purpose. This module tests whether recursive damping can be connected to recombination timing, visibility functions, and CMB-observable structure.

Role in the second preprint. The second preprint gave the CMB module a central position because the CMB is the main public empirical screen for early-universe models. The module is also a strong place to show how RFL memory becomes observable structure.

Upgrade path from the boundary ledgers. V2 and E2 upgrade this module by requiring CLASS/CAMB and public likelihood workflows. The CMB story must become an external run plan: compute observables, compare to Planck/ACT/SPT and BAO/SNe datasets, and report posterior shifts.

Current status. External-likelihood-ready, not externally validated. Carry late H0 proxy 70.173, sigma8/S8 proxy 0.808, scalar tilt proxy 0.964, and Hubble bridge fraction 49.517 percent. These must be labeled as proxies, not posterior means.

Boundary language. Says: V2 defines the cosmology likelihood interface. Doesn't say: RFC fits Planck or resolves H0 without likelihood runs.

5.8 Module 8: particle masses and Standard-Model-facing structure

Purpose. This module connects modal eigenfields to particle-sector tables while preserving the difference between internal projected units and public physical constants.

Role in the second preprint. The second preprint made particle masses one of the most ambitious module outputs. That ambition is useful but dangerous: it can overstate what an internal symbolic table proves.

Upgrade path from the boundary ledgers. The boundary-ledger materials split the old mass/constant claim into Y3, U/U2-U5, P2, N V2, and S/T. This is one of the strongest improvements. Y3 handles a frozen retest; U5 handles representation/charge/electroweak/flavor scaffold; P2 handles public constants; N V2 handles internal dimensional projection.

Current status. Bounded internal compatibility, not full Standard Model derivation. Carry Y3D score 0.900906, strict score 0.905671, checks 12/12, old theta13 error 28.23284 percent, new theta13 error 0.42315 percent, U5 9/9 pass, and P2 public-comparison boundary.

Boundary language. Says: RFC has a no-retune particle-sector scaffold. Doesn't say: RFC derives the full Standard Model.

5.9 Module 9: PMNS/CKM mixing and frozen particle-sector retest

Purpose. This module tests whether recursive eigen-overlap can represent flavor mixing once formula search is frozen.

Role in the second preprint. The second preprint included neutrino mixing and PMNS extraction as an operational domain. The fourth preprint later made the CKM θ_{13} gap explicit. That honesty is important and must be preserved.

Upgrade path from the boundary ledgers. Y3 is the clean boundary-ledger improvement. It freezes the selected Y3-C spin-foam/RFLMEM gate and retests without new search or retuning. The θ_{13} repair becomes a responsible case study in how to improve a module without hiding the old failure.

Current status. Frozen retest pass with bounded claim. Carry before/after θ_{13} errors: 28.23284 percent to 0.42315 percent, improvement 27.80969, Y3D score 0.900906, strict score 0.905671, checks 12/12.

Boundary language. Says: Y3 repairs a frozen internal mixing gap. Doesn't say: it proves the CKM/PMNS sector from first principles.

5.10 Module 10: decoherence, observer branch separation, and neural signatures

Purpose. This module studies branch separation, recursive identity, and possible neural signatures as measurement targets rather than metaphysical conclusions.

Role in the second preprint. The first and second preprints had strong observer language. It gave the theory a wide explanatory scope, but it also created overclaim risk if observer fields were presented as consciousness proofs.

Upgrade path from the boundary ledgers. Z2 fixes the tone. It converts observer/neural content into a public-data analysis protocol: EEG/MEG/iEEG targets, preprocessing, avalanche observables, fractal/recursive signatures, recurrence, phase coherence, and entropy measures.

Current status. External-analysis-ready, not consciousness validation. Carry observer branching score $0.75\text{e-}5$, branch separation final $0.14\text{e-}5$, decoherence proxy $0.57\text{e-}7$, fractal coherence proxy 0.812, recursive stability proxy 0.944.

Boundary language. Says: RFC defines an observer/neural measurement harness. Doesn't say: RFC validates consciousness or observer physics through EEG.

5.11 Current no-retune equation panels for Section 5

The module subsections above describe the architecture in prose. This subsection makes the shared mathematical form explicit. The goal is not to add ten separate fitted models. The goal is to show that each module is a modal projection of the same frozen packet:

$$\text{CIF modal source} \longrightarrow \text{QV compression/damping} \longrightarrow \text{RFL stabilized output}.$$

Only the modal basis changes from one lane to another. The no-retune packet is held fixed:

$$\mathcal{P}_G = \left\{ \delta = 4.6692, L = 60, \alpha = \frac{\ln \delta}{L} = 0.0256831, k = 2, \nu = 2\delta^{-4} = 0.00420784, \right. \\ \left. \epsilon = \alpha\nu = 1.08071 \times 10^{-4}, \lambda_\Delta = 0.489442, n_c = 18, n_\star = 40 \right\}.$$

The shared recursive kernel is written

$$K_m(t; n) = \lambda_\Delta \sum_{j=1}^n \delta^{-j} f_{m,j}(t) e^{-\alpha j t}, \quad n \in \{n_c, n_\star\}.$$

Here $n_c = 18$ is the deterministic closure depth and $n_\star = 40$ is the full canonical projection depth. CIF supplies the available modal basis $f_{m,j}$, QV supplies the compression factor $\delta^{-j} e^{-\alpha j t}$, and RFL is the stabilized output read from K_m .

Module lane	Changing term	Fixed terms and status
Inflation / Big Implosion	$f_{\text{BI},j}, f_{\text{inf},j}$	Frozen packet; internal expansion and forcing proxy.
Thermal transitions	$\Theta_{\text{QCD}}, \Theta_{\text{EW}}$	Frozen packet; thermal gate, not full finite-temperature SM.
Defects / walls	X_{wall}	Frozen packet; kink profile, not abundance constraint.
PBH / overdensity	$\ddot{f}_{\text{PBH},j}$	Frozen packet; collapse-window proxy.
Neutrino / BBN	Θ_ν	Frozen packet; N_{eff} and light-element handoff.
CP / EDM	θ_{CP}	Frozen packet; bounded source and suppression lane.
CMB / visibility	$S_{\text{rec}}, g_{\text{rec}}$	Frozen packet; public likelihood target.
Masses / constants	$\psi_a, \mathcal{B}_m$	Internal/projected units only until SI bridge.
Mixing	M_{ab}	Frozen overlap scaffold; comparison target.
Observer / neural	D_{ab}, A_a	Signal-analysis harness, not consciousness validation.

Module 1: Big Implosion, recursive inflation, and reheating

$$\begin{aligned}
\Phi_{\text{BI}}(t) &= K_{\text{BI}}(t; n_\star), & f_{\text{BI},j}(t) &= \cos(jt + \varphi_{\text{BI}}). \\
\Lambda_{\text{BI}}(t) &= \epsilon \Phi_{\text{BI}}(t) e^{-\alpha t}. \\
\phi_{\text{rec}}(t) &= K_{\text{inf}}(t; n_\star), & a_{\text{rec}}(t) &= \exp[\phi_{\text{rec}}(t)]. \\
H_{\text{rec}}(t) &= \dot{\phi}_{\text{rec}}(t), & \varepsilon_{\text{rec}}(t) &= \frac{1}{2} \dot{\phi}_{\text{rec}}(t)^2.
\end{aligned}$$

Boundary. These equations define an internal recursive inflation and Big Implosion lane. They do not replace a full perturbation-spectrum calculation until the lane is run against public CMB and large-scale-structure likelihoods.

Module 2: QCD and electroweak thermal transitions

$$\begin{aligned}
T_{\text{rec}}(t) &= T_0 e^{-\beta t}. \\
\Theta_x(t) &= \frac{1}{2} [1 + \tanh(\sigma_x [T_x - T_{\text{rec}}(t)])], & x &\in \{\text{QCD}, \text{EW}\}. \\
K_{\text{th}}(t; n_\star) &= \lambda_\Delta \sum_{j=1}^{n_\star} \delta^{-j} \Theta_{\text{QCD}}(t) \Theta_{\text{EW}}(t) \cos(jt) e^{-\alpha j t}.
\end{aligned}$$

Boundary. The gates identify where recursive structure should be tested across thermal history. They do not by themselves compute a full finite-temperature Standard Model effective potential.

Reader's note. These two modules begin the same dependency chain: the Big Implosion supplies the first compression lane, and the thermal gates identify where early recursive structure can pass into phase-transition targets.

Module 3: Domain walls and defect/kink geometry

$$\begin{aligned}
X_{\text{wall}}(t) &= K_{\text{wall}}(t; n_{\star}), & f_{\text{wall},j}(t) &= \sin(jt + \varphi_{\text{wall}}). \\
\Theta_{\text{wall}}(x, t) &= \pi \tanh[\kappa(x - x_0 - AX_{\text{wall}}(t))]. \\
\rho_{\text{wall}}(x, t) &= |\partial_x \Theta_{\text{wall}}(x, t)|^2.
\end{aligned}$$

Boundary. This gives the recursive defect profile used by the model. A physical defect abundance claim would require a full thermal-history and network-evolution calculation.

Module 4: Primordial black-hole and overdensity channel

$$\begin{aligned}
f_{\text{PBH},j}(t) &= \frac{\sin(jt)}{j}. \\
\rho_{\text{rec}}(t) &= \sum_{j=1}^{n_{\star}} \left[\delta^{-j} e^{-\alpha j t} \frac{d^2}{dt^2} f_{\text{PBH},j}(t) \right]^2. \\
\mathbb{I}_{\text{PBH}}(t) &= \mathbf{1}[\rho_{\text{rec}}(t) > \rho_c], & \lambda_{\text{rec}}(t) &= \frac{1}{T} \ln \left| \frac{\partial \psi_t}{\partial \psi_0} \right|.
\end{aligned}$$

Boundary. The threshold marks an internal collapse-window proxy. It is not a PBH mass function or observational abundance constraint until propagated through a full cosmological PBH pipeline.

Cycle role. Modules 3 and 4 are the first places where recursive compression looks spatial: one stabilizes as a defect profile, the other as a collapse-window proxy.

Module 5: Neutrino background and BBN interface

$$\begin{aligned}\Theta_\nu(t) &= [1 + \exp(\sigma_\nu[T_{\text{rec}}(t) - T_{\nu,\text{dec}}])]^{-1}. \\ K_\nu(t; n_\star) &= \lambda_\Delta \sum_{j=1}^{n_\star} \delta^{-j} \Theta_\nu(t) \cos(jt + \varphi_\nu) e^{-\alpha j t}. \\ N_{\text{eff}}^{\text{RFC}}(t) &= 3.046 \Theta_\nu(t) + \epsilon K_\nu(t; n_\star). \\ Y_p^{\text{RFC}} &= Y_p^{(0)} + \Delta Y_p[K_\nu, \Gamma_\zeta].\end{aligned}$$

Boundary. This lane defines how RFC enters neutrino decoupling and light-element windows. It does not claim to solve lithium or replace public BBN reaction-network calculations.

Module 6: CP phase, EDM windows, and baryogenesis

$$\begin{aligned}\theta_{\text{CP}}(t) &= \epsilon K_{\text{CP}}(t; n_\star). \\ S_{\text{CP}}(t) &= e^{-E_{\text{sph}}/T_{\text{rec}}(t)} \sin[\theta_{\text{CP}}(t)]. \\ \mu_B^{\text{RFC}}(t) &= \gamma_{\text{CP}} \frac{dS_{\text{CP}}(t)}{dt}. \\ d_f^{\text{RFC}} &= \mathcal{C}_f \theta_{\text{CP}} \mathcal{S}_{\text{fractal}}(n_\star), \quad \mathcal{S}_{\text{fractal}}(n) = \delta^{-2n}.\end{aligned}$$

Boundary. The CP lane supplies a bounded source and suppression structure. It does not become a physical EDM or baryogenesis result until mapped through EFT coefficients, transport, and experimental bounds.

Module 7: CMB visibility, recombination, and transparency

$$p_j(t; n) = \frac{\delta^{-j} e^{-\alpha j t}}{\sum_{k=1}^n \delta^{-k} e^{-\alpha k t}}.$$

$$S_{\text{rec}}(t; n) = - \sum_{j=1}^n p_j(t; n) \ln p_j(t; n).$$

$$\tau_{\text{rec}}(t) = e^{-S_{\text{rec}}(t; n_*)}, \quad g_{\text{rec}}(t) = -\dot{\tau}_{\text{rec}}(t) e^{-\tau_{\text{rec}}(t)}.$$

$$C_{\ell}^{\text{RFC}} = C_{\ell}^{\Lambda\text{CDM}} \left[1 + A_{\text{rec}} \cos \left(\frac{2\pi\ell}{\delta} + \varphi_{\text{rec}} \right) \right].$$

Boundary. The entropy and visibility equations define a structured test target. They do not claim a CMB fit until evaluated through CLASS/CAMB and public likelihoods.

Module 8: Particle masses and Standard-Model-facing structure

$$\psi_a(t) = K_a(t; n_*), \quad f_{a,j}(t) = \cos(jt + \chi_a).$$

$$m_a^{\text{int}} = \eta_m \langle |\psi_a(t)|^2 \rangle_T^{1/2}.$$

$$m_a^{\text{proj}} = \mathcal{B}_m(\mathcal{P}_G, \mathcal{R}, \mathcal{N}) m_a^{\text{int}}, \quad \mathcal{B}_m \neq \text{SI calibration}.$$

Boundary. RFC can define internal symbolic and projected masses, but public particle masses require an external calibration and uncertainty bridge.

Module 9: PMNS/CKM mixing and frozen particle-sector retest

$$\Psi_a(t) = K_a(t; n_\star), \quad f_{a,j}(t) = \sin(jt + \chi_a).$$

$$\langle \Psi_a, \Psi_b \rangle_T = \frac{1}{T} \int_0^T \Psi_a(t) \Psi_b(t) dt.$$

$$M_{ab} = \frac{\langle \Psi_a, \Psi_b \rangle_T}{\|\Psi_a\|_T \|\Psi_b\|_T}, \quad U_{\text{rec}} = \text{polar}(M).$$

Boundary. This is a frozen internal mixing scaffold. It may be compared to CKM/PMNS data, but it should not be described as a full Standard Model derivation.

Module 10: Observer branches, decoherence, and neural signals

$$\Psi_a(t) = K_a(t; n_\star), \quad \Psi_{\text{obs}}(t; \xi) = K_{\text{obs}}(t; n_\star).$$

$$D_{ab}(T) = \int_0^T |\Psi_a(t) - \Psi_b(t)| dt.$$

$$A_a = \frac{\langle \Psi_{\text{obs}}, \Psi_a \rangle_T}{\|\Psi_{\text{obs}}\|_T \|\Psi_a\|_T}, \quad \mathcal{D}_{\text{branch}} = 1 - \max_a A_a.$$

$$\frac{\partial^2 \phi}{\partial t^2} = \nabla^2 \phi - \lambda \phi + K_{\text{neural}}(x, y, t; n_\star).$$

Boundary. The observer and neural equations define signal-analysis targets. They do not validate consciousness, perception, or cognition without preregistered public EEG/MEG/iEEG analysis.

6 New boundary lanes beyond the module architecture

The boundary ledgers add several lanes that were not cleanly separated in the early preprints. These lanes should not be hidden in appendices only. They are part of the scientific maturation of RFC.

The equation panels just inserted make the boundary logic sharper. A module can have a well-defined mathematical lane and still require an external public-data run. A clean formula is not the same thing as a confirmed observable. For this reason, the lanes below are written as handoffs: they specify what must be compared, with what public data, using what public or reproducible code, before a stronger claim is allowed.

The strongest version of the paper is therefore not the most totalizing version. It is the version that distinguishes internal derivation, internal closure, proxy agreement, external-run readiness, and empirical validation. The next subsections use that vocabulary for the particle, cosmology, BBN, CP/EDM, gravitational-wave, observer, and early-universe boundaries.

6.1 Y3: a model repair done correctly

Y3 is the cleanest example of a responsible repair. The fourth preprint acknowledged that the Y2 particle-sector map was exploratory and that CKM θ_{13} remained too high. The boundary-ledger work freezes the mechanism and retests it without new search. The before/after result is stark: the error changes from 28.23284 percent to 0.42315 percent, with 12/12 checks passing.

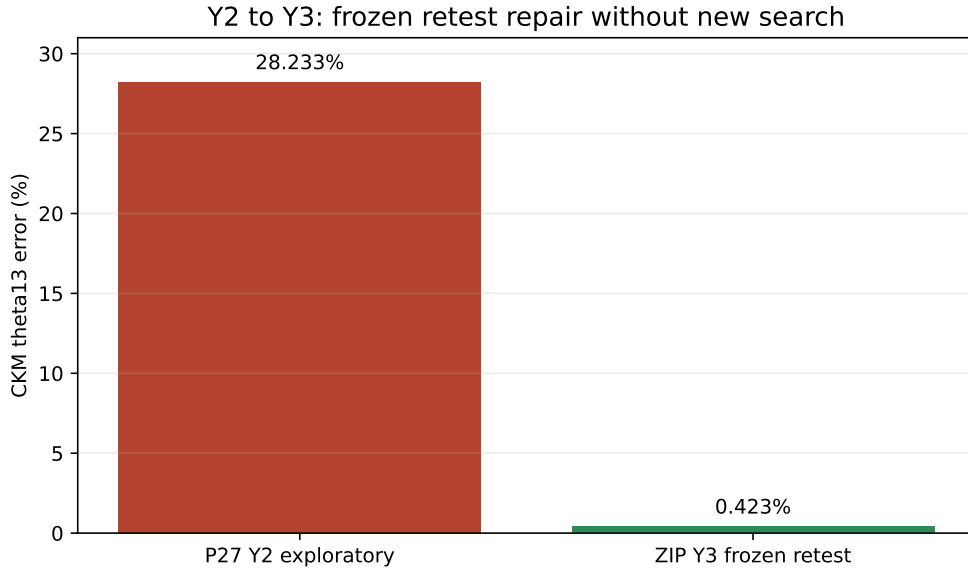


Figure 6: Y3 repairs a specific exploratory weakness under a frozen retest. The figure should be read as an internal repair, not as a full particle-physics derivation.

The right language is: the particle-sector retest is improved, frozen, and internally bounded. The wrong language is: RFC has derived the Standard Model.

6.2 U/U2-U5 and P2: separating scaffold from constants

The U sequence converts a broad particle claim into a scaffold. U5 includes charge quantization proxy, hypercharge branch lock, sterile/Higgs locks, anomaly scaffold, chirality orientation repair, weak-isospin generator carry, electroweak closure, gauge/Higgs mass carry, and flavor-matrix consistency. Its ledger reports 9/9 pass. P2 is different. P2 is not the same as U5. It is the public constants and parameter-table boundary.

This separation is important enough to state as a rule: U5 may discuss representation, charge, electroweak, and flavor compatibility. P2 must handle CODATA, NIST, PDG, and covariance comparison. Neither should claim a complete derivation of public constants.

6.3 V2: precision cosmology boundary

V2 upgrades a compressed cosmology screen into a public-likelihood plan. It names the public data and software that matter: DESI BAO, Pantheon+, DES-SN, Planck, ACT/SPT, growth/lensing data, CLASS, CAMB, Cobaya, and MontePython. It carries internal quantities such as late H0 proxy 70.173, middle residual 0.173, and Hubble bridge fraction 49.517 percent.

The important sentence is: V2 is external-run-ready, not precision-cosmology-solved. A credible paper can include the proxy, but it must also state the likelihood outputs required: posterior chains, residuals against BAO/SNe/CMB, matter power spectra, growth observables, and model-comparison statistics.

6.4 W2: BBN and the lithium wall

W2 is stronger because it refuses to hide failure. It reports Y_p error 0.922 percent, D/H error 2.000 percent, He3 error 3.000 percent, and Li7 error 212.500 percent. The lithium problem remains a wall. The required suppression is about 68 percent, and required channel modulation is about 36.957 percent.

This is not a weakness if written correctly. It makes the paper credible. W2 says: the proxy and handoff close; the lithium problem does not. The external work is a public BBN code run with specified reaction channels, priors, and abundance comparisons.

6.5 X2: CP, EDM, and baryogenesis boundary

X2 prevents one of the most tempting overclaims. A small CP residual or EDM proxy is not a physical EDM prediction. A baryon eta proxy is not a baryogenesis proof. X2 therefore names the missing physics: operator basis, Wilson coefficients, physical conversion factors, transport equations, washout modeling, and external bounds.

The analysis should use X2 to show maturity. A model that names its missing EFT layer is stronger than a model that hides it.

6.6 GW2: ringdown and echo window

GW2 carries a quality factor proxy 7.427, echo delay proxy 1.541, echo spacing proxy 0.103, and rebirth boundary gap 0.195. These are useful waveform-handoff quantities. They are not detections. The public path runs through GWOSC data, waveform generation, ringdown likelihoods, Bayesian model comparison, and residual analysis.

6.7 Z2: observer and neural analysis

Z2 turns observer language into a measurement protocol. It specifies public EEG/MEG/iEEG data targets, preprocessing, avalanche statistics, fractal dimension, Hurst exponents, spectral slopes, recurrence metrics, phase coherence, and entropy measures. It carries internal proxies such as fractal coherence 0.812 and recursive stability 0.944.

The section must avoid consciousness language except as motivation. The scientific claim is only that RFC defines a candidate observer/neural signature harness.

6.8 E2: early-universe relic and transition registry

E2 recovers much of the earlier explanatory scope while preserving claim boundaries. It includes inflation/reheating, QCD/EW transitions, domain walls, PBHs, relics, recombination, public sources, and external task packets. The point of E2 is not to prove all early-universe phenomena. It is to prevent the paper from scattering early-universe claims across unrelated sections.

Downstream lane status: credible boundaries before claims



Figure 7: Downstream boundary lanes. A lane can be valuable even when it is not externally validated, if it specifies the public data, code, outputs, and blocked claims.

7 Quantitative spine and internal screen summary

This section collects the numerical quantities that should appear in the main text. The table is not a proof. It is a compact statement of what the current RFC rebuild is allowed to carry into downstream modules.

Table 3: Core internal quantities and their allowed interpretation.

Quantity	Value	Allowed interpretation
Module G packet	$\delta = 4.6692, L = 60, \alpha = 0.0256831$	Frozen deterministic source packet, no empirical targets
ν, ϵ	0.00420784, 0.000108071	Internal phase/damping derived from G packet
λ_N	0.489442	Internal normalized closure value
Module R residual improvement	0.418175	Source-coupled closure improvement, not an external fit
Module R source-coupled score	0.576064	Residual score under V2 standardization
Module R score V2	0.726401	Interpretability-oriented closure summary
QV grouped power	0.794508	QV-dominant source closure
Dark-kernel bridge power	0.19036	Bridge between QV-dominant flow and RFL residual structure
CP residual	3.59×10^{-10}	Internal stability indicator
Module N energy/time factors	0.28443, 1.54099	Internal dimensional projection bridge
Projected cycle time	92.459	Internal projected unit
Y3 theta13 repair	28.23284% to 0.42315%	Frozen internal retest improvement
W2 Yp/D/H/He3 errors	0.922%, 2.000%, 3.000%	Light-abundance proxy window
W2 Li7 error	212.500%	Explicit unresolved lithium wall
V2 late H0 proxy	70.173	Cosmology handoff proxy, not a posterior
X2 eta10 proxy	6.068	Source-window proxy, not baryogenesis proof

Quantity	Value	Allowed interpretation
E2 scalar tilt proxy	0.964	External cosmology target proxy
GW2 Q/delay/spacing proxies	7.427, 1.541, 0.103	Waveform-boundary quantities, not detections
Z2 fractal coherence	0.812	Neural-analysis target proxy

8 Public-data and public-code handoff

A scientific preprint must explain how the internal structure can be tested. The following handoffs should be treated as required future work, not as already completed validation.

8.1 Cosmology handoff

The V2 lane requires a public cosmology pipeline. The minimum viable pipeline is: implement the RFC background or effective source in CLASS or CAMB; compute distances, Hubble history, CMB spectra, matter power, and growth quantities; sample or compare using Cobaya or MontePython; include Planck, DESI BAO, Pantheon+ or supernova data, and growth/lensing where appropriate. The output should include posterior chains and residual plots, not only point comparisons.

8.2 BBN handoff

The W2 lane requires PARthENoPE, PRIMAT, AlterBBN, or equivalent BBN code comparison. The lithium wall must be carried openly. A credible run must report Y_p , D/H , $He3$, $Li7$, reaction-channel sensitivity, and whether any $Li7$ improvement damages the already acceptable light-element windows.

8.3 Particle and constants handoff

The Y3/U5/P2/S/T lanes require PDG, CODATA/NIST, BIPM, NuFIT, and covariance-aware comparisons. The internal projection bridge cannot be treated as a constants derivation. Dimensionless ratios and one-anchor calibration are the correct public interface.

8.4 CP/EDM/baryogenesis handoff

X2 requires EFT conversion. Internal EDM-like proxies become meaningful only after specifying operators, Wilson coefficients, hadronic/nuclear/atomic conversion, and the relevant experimental bounds. Baryogenesis requires source terms, transport, washout, and cosmological consistency with observed baryon asymmetry.

8.5 Gravitational-wave and neural handoffs

GW2 requires waveform likelihoods against public GW data. Z2 requires public neurodata preprocessing and statistical tests. In both cases the preprint should present measurement plans and internal target signatures, not detections.

9 Figures, tables, and page architecture

The visual doctrine is simple: diagrams teach dependency; tables decide status; plots show what changed. Public images may be used later, but this draft uses original vector-style figures to avoid

licensing and endorsement problems.

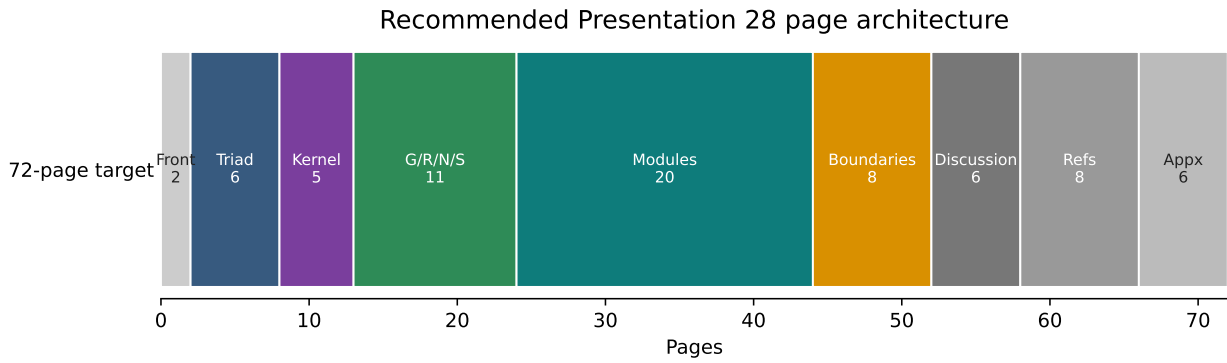


Figure 8: Recommended page architecture from the page-map analysis. The target is roughly 72 pages, with the main body carrying the triad, kernel, deterministic closure, module architecture, and boundaries before references and appendices.

10 Falsifiability and blocked claims

RFC becomes more credible when it states what would count against it. The following blocked claims are not stylistic caveats. They are necessary scientific boundaries.

1. RFC does not currently claim to derive public SI constants from first principles.
2. RFC does not claim to solve the full Standard Model.
3. RFC does not claim to solve precision cosmology or the Hubble tension without public likelihoods.
4. RFC does not claim to solve Li7.
5. RFC does not claim to predict physical EDMs without EFT conversion.
6. RFC does not claim to prove baryogenesis without transport and washout calculations.
7. RFC does not claim to detect gravitational-wave echoes.
8. RFC does not claim to validate consciousness or neural identity.
9. RFC does not claim that internal projected units are public measurements.
10. RFC does not claim that proxy screens are empirical validation.

A future external run can change some of these statuses. Until then, the block remains part of the theory's scientific presentation.

11 Discussion

The main conceptual result of the rebuild is a change in order. Earlier RFC documents often moved from broad explanatory scope into modules, then into validation language. this preprint reverses the risky part of that pattern. It begins with the triad, builds the kernel, freezes the source packet, audits closure, defines projection bridges, and only then enters downstream modules.

This change improves the theory in three ways. First, it makes the triad operational. CIF, QV, and RFL are no longer repeated names; they are source, operation, and stabilized output. Second, it prevents downstream retuning. The same packet must survive every module. Third, it changes the tone of validation. A module can be valuable as a handoff even when it is not yet externally confirmed.

The strongest version of RFC is therefore not the version that says it has solved everything. The strongest version says exactly what is internally closed, what is a proxy, what has a public-data route, and what remains blocked. That version is more scientific and more durable.

12 The collapse-rebirth cycle across the module system

The module architecture is not just a list of tests. Read in order, it describes one cycle: a collapse of modal possibility into stabilized structure, followed by the return of that structure as recursive memory. The Big Implosion is the first large-scale instance of this cycle. It is the moment where QV acts on CIF with maximum compressive intensity and forces a branch of possibility into RFL structure. The later modules are not separate stories added after the fact. They are phase-resolved expressions of the same event becoming physically legible.

In this paper, *collapse* does not mean the end of existence, and *rebirth* does not mean a mystical restart. Collapse means source compression, branch selection, damping, and asymmetry formation. Rebirth means stabilized recursive output: the selected branch becomes geometry, relic structure, particles, radiation, memory, observer separation, and a new source condition for subsequent closure. The cycle can be summarized as

$$\text{Big Implosion} \rightarrow \text{QV}(\text{CIF}) \rightarrow K_{f_j} \rightarrow \text{RFL}_n \rightarrow M_{\text{rec}} \rightarrow \text{QV}(\text{CIF})_{n+1}.$$

Here, K_{f_j} is the recursive modal kernel and M_{rec} denotes the memory returned by the previous stabilized state. The scientific point is modest but important: every module is assigned a phase in the cycle, and the model should not claim more than that phase has earned.

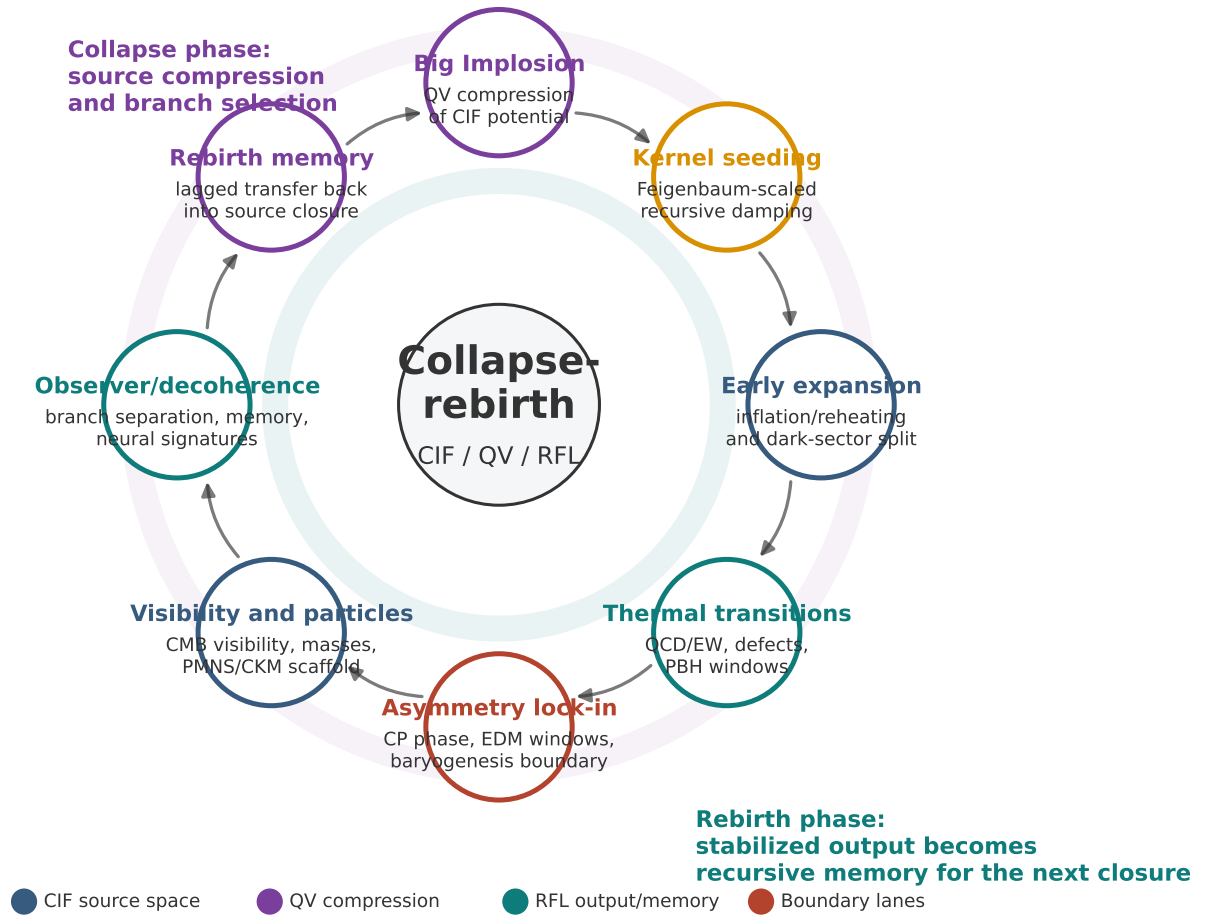


Figure 9: Collapse-rebirth cycle generated by the module system. The Big Implosion begins the cycle as QV compression of CIF potential. The kernel distributes that compression into domain-specific modal bases. The downstream modules then describe how expansion, transitions, asymmetry, visibility, particles, decoherence, and memory become stabilized RFL outputs. The return arrow is a memory return, not a completed claim of cyclic cosmology.

Cycle phase	Active modules	Scientific role	Boundary
Big Implosion	Triad, kernel, Module G	Establish the first compression of CIF source space by QV into a frozen, no-target source packet. This is the collapse seed.	Internal derivation; not external cosmological validation.
Kernel propagation	Recursive kernel, Feigenbaum scaling, entropy memory	Convert the first action into a reusable modal operator. The same skeleton can be projected into different bases without retuning the packet.	Shared skeleton, not a universal solution by itself.
Early rebirth	Inflation/reheating, dark-sector compression and tail	Treat expansion as the first RFL rebound after compression. Dark matter-like compression and dark-energy-like tail are read as distinct residues of the same forcing history.	Requires public cosmology likelihoods before observational claims.
Thermal differentiation	QCD/EW transitions, domain walls, PBH windows, neutrino background	The rebound becomes phase structure: transitions, defects, overdensities, decoupling, and relic populations.	Registry of windows until specialized solvers are run.
Asymmetry lock-in	CP phase, EDM windows, baryogenesis	Local branch bias becomes matter-facing asymmetry. The cycle selects a branch without claiming global asymmetry of the whole source space.	Needs EFT conversion, transport equations, washout tests, and experimental bounds.
Visibility and particle stabilization	CMB visibility, masses, PMNS/CKM, U5/P2	RFL output becomes measurable: radiation transparency, particle scaffolds, mixing structures, and constants-comparison tables.	Scaffolds and internal retests are not full Standard Model derivations.
Observer separation	Decoherence, observer branch separation, Z2	Stabilized branches become distinguishable histories. Neural and observer modules test whether recursive memory has measurable signatures.	No claim of consciousness validation; public data and preregistration required.
Memory return	Module R, Module N, S/T, LEG2/TK2	The stabilized output is audited as source-coupled residual structure, projected dimensionally, and returned as a bounded construction rule for the next revision.	Closure audit and manuscript discipline, not a proof of cyclic cosmology.

This cycle explains why the module order matters. Inflation, thermal transitions, CP asymmetry, CMB visibility, particle structure, observer separation, and legacy reconciliation are not independent decorations. They are successive readings of one recursive process under different observational projections. The collapse side asks how possibility is compressed and selected. The rebirth side asks what survives as structure, memory, and measurable residue.

The cycle also clarifies why the strongest claims remain blocked. A collapse-rebirth architecture can organize the modules, but the paper should not claim a completed cyclic cosmology until external runs show that the same frozen packet survives public data without retuning. The present status is stronger and narrower: RFC supplies a coherent cycle map, a frozen source packet, a source-coupled closure audit, and a set of public routes by which the cycle can be tested.

13 Conclusion

RFC is reconstructed here as a triad-derived recursive framework. CIF supplies modal possibility. QV supplies compression, damping, and selection. RFL supplies stabilized recursive output. The first action $QV(CIF) \rightarrow RFL$ becomes a universal kernel with domain-specific modal bases. Module G freezes the source packet without empirical targets. Module R V3 audits source-coupled closure. Module N V2 separates internal projected units from public constants. Modules S/T define the unit/coupling interface. Downstream lanes then become bounded tests rather than unguarded claims.

This is the right form for this preprint. It preserves the early explanatory power, retains the module

architecture, keeps the mathematical codex as support, and carries the discipline introduced by the boundary ledgers. The next scientific step is external execution: public cosmology likelihoods, public BBN runs, particle/constant comparisons, EFT/EDM conversion, waveform likelihoods, and public neural data analysis.

A Full boundary module ledger

Module	Function	Data to carry	Status	Boundary
G	Deterministic triadic closure	delta=4.6692; L=60; alpha=0.0256831; nu=0.00420784; epsilon=0.000108071; lambda_N=0.489442	Frozen internal packet	No empirical targets; no MCMC/NUTS
R V3	Source-coupled triad closure	QV core=0.794508; dark bridge=0.19036; residual improvement=0.418175	Closure audit pass	Not a fit to external data
N V2	Dimensional projection	energy factor=0.28443; time factor=1.54099; identity residuals near zero	Internal projection pass	Not SI constants
S/T	SI and coupling bridge	one-anchor bridge; dimensionless coupling map	Bridge scaffold	External calibration required
Y3	Particle-sector frozen retest	theta13 error 28.23284% -i, 0.42315%; checks 12/12	Frozen pass	Not full SM derivation
U5	SM-facing scaffold	charge/hypercharge/EW/flavor ledger; 9/9 pass	Bounded pass	Representation scaffold only
P2	Constants boundary	CODATA/PDG/NIST public-comparison handoff	Handoff ready	No constants derivation
V2	Precision cosmology	H0 proxy 70.173; CLASS/CAMB/Cobaya handoff	External-run-ready	No BAO/SNe/CMB likelihood pass yet
W2	BBN/Li7 boundary	Yp err 0.922%; D/H 2%; He3 3%; Li7 212.5%	Proxy pass with wall	Li7 unresolved
X2	CP/EDM/baryogenesis	J proxy 0.32e-4; eta10 6.068; EDM windows	Source-window pass	No physical EDM/transport proof
Z2	Observer/neural/EEG	fractal coherence 0.812; recursive stability 0.944	Protocol ready	No consciousness validation
GW2	Ringdown/waveform boundary	Q proxy 7.427; delay 1.541; spacing 0.103	Waveform-ready	No detection claim
E2	Early-universe relics	tilt 0.964; r 0.0017; PBH fraction 3.4e-5	Thermal-history ready	External solvers needed
LEG2	Legacy reconciliation	legacy claim and module reconciliation	Narrative pass	No empirical claim
TK2	Manuscript assembly	dependency order; page map; figure/table plan	Assembly controller	Not a science module

B Symbol glossary

Symbol	Meaning in this draft
CIF	Cosmic Infinite Field; modal source space or possibility domain
QV	Quantum Vacuum; compression, damping, entropy-gradient, and selection operation
RFL	Recursive Fractal Lattice; stabilized recursive output, memory, geometry, observability
δ	Feigenbaum scaling constant used as recursive-bifurcation scale
L	Cycle length in the frozen packet
α	Damping rate $\log(\delta)/L$
ν	Phase-depth derived quantity $2\delta^{-4}$
ϵ	Small composite phase/damping quantity $\alpha\nu$
λ_N	Normalized internal closure quantity
$K_f(t)$	Universal recursive kernel with modal basis $f_j(t)$
$f_j(t)$	Domain-specific modal function at recursion depth j
G	Deterministic triadic closure module
R V3	Source-coupled triad closure audit
N V2	Dimension-aware projection bridge
S/T	SI bridge and dimensionless coupling interface
Y3	Frozen particle-sector retest
U5	Standard-Model-facing bounded scaffold
P2	Public constants and parameter comparison boundary

Symbol	Meaning in this draft
V2	Precision-cosmology external-run boundary
W2	BBN and Li7 boundary
X2	CP/EDM/baryogenesis source-window boundary
Z2	Observer/neural/EEG external-analysis boundary
GW2	Gravitational-wave waveform/ringdown boundary
E2	Early-universe relic and transition boundary

C External-run registry

Lane	Required public data	Required public software or method	Required output
V2	Planck, DESI BAO, Pantheon+, DES-SN, ACT/SPT, growth/lensing	CLASS/CAMB with Cobaya/MontePython	Posterior chains, residuals, model comparison
W2	Observed primordial Y_p , D/H, He3, Li7	ParthENoPE, PRIMAT, AlterBBN	Abundance table, channel sensitivities, Li7 diagnosis
X2	EDM bounds, baryon asymmetry, CKM/PMNS inputs	EFT/Wilson conversion, transport equations	Physical EDM predictions, washout/source analysis
P2	CODATA/NIST, BIPM, PDG, NuFIT	Dimensionless-ratio comparison, covariance accounting	Constants/ratios table with uncertainty propagation
GW2	GWOSC strain data and catalogs	Waveform/ringdown likelihoods, Bayesian model comparison	Echo/ringdown residual statistics, null tests
Z2	Open EEG/MEG/iEEG datasets	MNE-Python, avalanche and fractal statistics	Exponent tables, recurrence/phase/entropy metrics
E2	CMB, relic, thermal transition, PBH constraints	Inflation/reheating/thermal-history solvers	Thermal history comparison packet
QG	Spin-foam and quantum-geometry benchmarks	State-sum/graph audits	Finite-suite consistency diagnostics

D Module-specific boundary dossier

This appendix gives a module-by-module boundary dossier. Each lane states what it carries internally, what public comparison it still requires, and what would count as a failure. The purpose is not to add more claims; it is to prevent internal closure, proxies, handoffs, and validations from being mixed together.

D.1 G: deterministic triadic closure

Preprint function. The G lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that the source packet must be generated before any empirical comparison is made. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is recomputing α , ν , ϵ , λ_N , n -closure, and n -full from the same deterministic rules. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if a downstream module alters any frozen number to improve agreement. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.2 R V3: source-coupled closure audit

Preprint function. The R V3 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that closure must be read as a source decomposition, not as a generic residual score. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is reproducing raw residual, source-coupled residual, source power fractions, lagged memory transfer, and CP residual without retuning. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if the source groups change after a downstream target is introduced. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.3 N V2: dimension-aware projection bridge

Preprint function. The N V2 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that internal symbolic values must not be confused with public SI

constants. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is checking mass-squared, inverse-energy, and cycle identities under the energy and time factors. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if a projected internal value is presented as a measured constant. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.4 S/T: unit and dimensionless coupling interface

Preprint function. The S/T lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that external comparison must pass through either a declared anchor or a dimensionless ratio. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is recording the anchor, the dimensionless observable, the public reference value, and the uncertainty path. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if the paper reports unanchored dimensional agreement. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.5 Y3: frozen particle-sector retest

Preprint function. The Y3 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that a known exploratory weakness must be repaired only under a frozen retest. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is showing the Y2 theta13 gap, the frozen Y3 mechanism, and the final error after retest. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if the formula is searched again after seeing the error. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.6 U5: Standard-Model-facing scaffold

Preprint function. The U5 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that representation and charge compatibility can be audited without claiming a full derivation of the Standard Model. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is listing charge, hypercharge, chirality, electroweak, Higgs, and flavor checks with pass/fail status. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if the section implies that gauge theory and all particle masses have been derived. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.7 P2: constants and parameter-table boundary

Preprint function. The P2 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that public constants require CODATA, NIST, PDG, BIPM, and covariance-aware comparison. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is building a table of constants, units, uncertainty, internal RFC quantity, and status label. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if a symbolic number is treated as a CODATA value. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.8 V2: precision cosmology external-likelihood lane

Preprint function. The V2 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that cosmological credibility requires public likelihoods, not compressed proxies alone. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is running CLASS or CAMB with BAO, SNe, CMB, growth, and matter-power comparisons. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if the module claims to solve precision cosmology before a likelihood run exists. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.9 W2: BBN and Li7 boundary

Preprint function. The W2 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that a successful light-element proxy can coexist with an unresolved lithium wall. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is reporting Y_p , D/H, He3, Li7, channel sensitivity, and public BBN-code outputs. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if Li7 is hidden or described as solved. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.10 X2: CP/EDM/baryogenesis source-window

Preprint function. The X2 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that dimensionless CP and EDM proxies must be converted through EFT and transport physics before physical claims. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is specifying operators, Wilson coefficients, conversion factors, bounds, sources, washout, and baryon eta. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if a small proxy is called a physical EDM prediction. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.11 GW2: gravitational-wave waveform boundary

Preprint function. The GW2 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that ringdown and echo proxies are waveform targets, not detections. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is using public strain data, waveform templates, ringdown likelihoods, null tests, and Bayesian model comparison. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if echo language is presented as observational evidence. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.12 Z2: observer, neural, and EEG boundary

Preprint function. The Z2 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that observer language must be turned into testable signal statistics. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for

explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is processing EEG/MEG/iEEG data for avalanche exponents, Hurst scaling, spectral slope, recurrence, phase coherence, and entropy. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if the section claims validation of consciousness. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.13 E2: early-universe relic and transition registry

Preprint function. The E2 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that the early-universe narrative must be gathered into a single thermal-history work plan. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is connecting inflation, reheating, QCD/EW transitions, defects, PBHs, relics, and recombination to public constraints. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if the paper claims all early-universe relics are solved. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.14 QG/QG2/J2: quantum-geometry and spin-foam audit

Preprint function. The QG/QG2/J2 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that quantum-geometry compatibility requires finite-suite audits and clear separation from a full quantum-gravity proof. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is recording graph/state choices, weights, identity tests, coherence behavior, and boundary conditions. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if the section claims RFC has completed quantum gravity. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.15 LEG2: legacy reconciliation

Preprint function. The LEG2 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that older explanatory material must be preserved in content but superseded in claim status. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is mapping each legacy claim to retained, translated, bounded, or deprecated status. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if older overclaims are silently carried into the new paper. This failure condition should remain explicit so that readers know how the claim could be wrong.

D.16 TK2: manuscript assembly controller

Preprint function. The TK2 lane exists to keep one part of the RFC rebuild from doing too much work. Its guiding principle is that the paper must be built in dependency order and not as a

miscellaneous bundle of modules. In this preprint this lane should appear after the reader understands the triad and the frozen packet. It should never be introduced as an isolated result.

Evidence and implementation role. Use the latest values and status labels from the boundary-ledger materials first. Use the fourth preprint only for the canonical rebuild context. Use the second preprint for module flow where that flow remains scientifically useful. Use the first preprint only for explanatory language, and translate that language into source, operator, output, proxy, or boundary terminology.

Scientific test. The minimum scientific test for this lane is checking title, page map, figure order, module order, reference map, claim labels, and blocked-claim registry. This test should be stated as a reproducible task. If it cannot be performed in the main text, it must be listed in the external-run registry with data, code, output, and pass/fail criteria.

Main-text wording. The main text should use active, bounded language: the lane defines, carries, audits, tests, or hands off. It should avoid totalizing verbs such as proves, explains all, completes, or derives unless the claim is mathematically demonstrated inside the section.

Failure condition. The lane fails its current claim status if the paper repeats the fourth preprint dryness or the second preprint overclaim. This failure condition should remain explicit so that readers know how the claim could be wrong.

E Scientific-language conversion table

The early RFC documents used strong symbolic language. this preprint must preserve the insight while changing the grammar. The following table converts high-risk phrasing into scientific phrasing.

Legacy phrase	Scientific conversion	Reason
CIF is infinite possibility	CIF is the modal source space	Keeps the role without meta-physical inflation
QV gives birth to structure	QV supplies compression, damping, and selection	Turns origin language into an operation
RFL is the child of emergence	RFL is stabilized recursive output and memory	Preserves emergence while naming observables
Big Implosion creates the universe	The Big Implosion is RFC’s first compression ansatz	Prevents cosmological over-claim
Entropy is memory, not disorder	RFC adds a recursive-memory interpretation to entropy gradients	Does not deny thermodynamics
RFC derives dark matter	RFC defines a dark-kernel compression proxy and handoff	Blocks premature dark-matter claims
RFC derives dark energy	RFC defines a dissipative-tail proxy and external cosmology lane	Requires V2 likelihoods
RFC solves baryogenesis	RFC defines a CP source-window and baryogenesis handoff	Requires transport and washout physics
RFC derives particle masses	RFC defines an internal particle-sector scaffold	Requires PDG and unit bridge comparison
RFC predicts EEG signatures	RFC defines a neural/observer measurement harness	Requires public neurodata analysis
RFC proves many worlds	RFC uses branch-separation/decoherence language as a module target	Blocks metaphysical conclusion
RFC proves quantum gravity	RFC tests compatibility through finite spin-foam/geometry audits	Avoids full proof claim

F Detailed external-run protocols

Each external lane should have the same structure: target question, public data, public code, internal RFC inputs, output products, and claim decision. This prevents the paper from sounding like it is asking readers to trust an internal notebook.

F.1 V2 protocol: cosmology

Target question: can the frozen RFC packet, once projected through the V2 cosmology lane, improve or remain compatible with public distance, CMB, and growth data? Public data: DESI BAO, Pantheon+ or equivalent SNe, Planck and other CMB likelihoods, and growth/lensing samples. Public code: CLASS or CAMB for observables, Cobaya or MontePython for sampling. RFC inputs: frozen G packet, V2 background prescription, dark-tail or compression terms only as declared. Outputs: posterior chains, best-fit comparison, residuals, model-comparison statistics, and public scripts.

F.2 W2 protocol: BBN

Target question: can the RFC source-window reproduce Y_p , D/H, and He3 without worsening Li7? Public data: primordial abundance compilations. Public code: PRIMAT, PArthENoPE, AlterBBN, or equivalent. RFC inputs: W2 source and damping proxies. Outputs: abundance table, reaction-channel sensitivities, uncertainty propagation, and a clear lithium-wall conclusion.

F.3 X2 protocol: EDM and baryogenesis

Target question: can the internal CP source be converted into physical EDM and baryogenesis observables? Public data: electron EDM, neutron EDM, atomic/molecular EDM bounds, baryon asymmetry, CKM/PMNS references. Public method: EFT operator basis, Wilson coefficients, hadronic/nuclear conversion, and transport equations. Outputs: physical EDM values, exclusion or allowed windows, eta comparison, washout sensitivity, and null results where necessary.

F.4 Y3/U5/P2 protocol: particle sector and constants

Target question: can the frozen particle-sector retest and representation scaffold survive public particle-data comparison? Public data: PDG, NuFIT, CODATA/NIST, BIPM. Method: dimensionless ratios, covariance accounting, explicit unit bridge. Outputs: CKM/PMNS error table, charge/hypercharge scaffold table, constants comparison table, and a list of values that remain internal-only.

F.5 GW2 protocol: waveform analysis

Target question: do RFC ringdown/echo-window proxies generate any statistically meaningful residual structure in public GW data? Public data: GWOSC strain and event catalogs. Public method: waveform generation, ringdown fitting, residual analysis, Bayesian model comparison, injection studies. Outputs: posterior odds, null tests, event-by-event residuals, and a statement that no detection is claimed unless the external analysis passes.

F.6 Z2 protocol: neural and observer signatures

Target question: do RFC recursive signatures appear in public neurophysiology data under a pre-registered analysis? Public data: EEG, MEG, or iEEG repositories with suitable sampling and metadata. Method: preprocessing, artifact rejection, avalanche extraction, power-law fitting, Hurst analysis, fractal dimension, recurrence, phase coherence, entropy measures. Outputs: tables of exponents, confidence intervals, null comparisons, and robustness checks.

G Visual and table production plan

The final PDF should be visually strong without sacrificing scientific discipline. The visual plan should follow four rules. First, every figure answers one question. Second, every table decides a claim status. Third, public images should be used sparingly and credited. Fourth, original vector diagrams should carry the RFC-specific argument.

Visual	Question answered	Required content
Triad roles	What are CIF, QV, and RFL scientifically?	Three nodes, first action, role labels
Dependency spine	Where do data enter?	Triad, kernel, G, R, N, S/T, ledgers

Visual	Question answered	Required content
Kernel diagram	How can one kernel support many modules?	Kernel equation plus modal outputs
Module matrix	Which triad role dominates each module?	Modules vs CIF/QV/RFL/status columns
Module R chart	What drives closure?	Source-power fractions
Y3 repair chart	What was repaired?	Old vs new theta13 error
Boundary dash-board	Which modules are external-run ready?	Lane status and blocked claims
Public-data map	What outside data are required?	Dataset, code, output, decision
Reference map	Which references support which module?	Reference cards by module
Page architecture	How is the preprint organized?	Page budget and section distribution

H Expanded module dossier

This appendix gives a compact module dossier. Each entry names the scientific role, evidence role, figure role, table role, reference role, and falsifier role. It is included so the reader can see how the same claim discipline is meant to travel across the full framework.

H.1 Triad foundation: CIF/QV/RFL definitions

Scientific role. The preprint must first define source space, compression operator, and stabilized memory. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is not a single measurement; it is failure of the triadic roles to map consistently onto the kernel and modules. This statement should remain visible. It is better for

the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.2 Kernel: recursive modal operator

Scientific role. The kernel is the bridge between the triad and all downstream modules. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is module-specific: if modal functions cannot be specified without retuning, the kernel is not doing the work claimed. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.3 G: deterministic closure

Scientific role. The packet must be derived once and carried everywhere. Each section should begin with the physical question, not with the result. The reader should know what the module is

trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is any need to alter the packet after looking at downstream targets. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.4 R V3: closure audit

Scientific role. R V3 explains which source groups carry residual structure. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is loss of source-coupled improvement or unstable group powers under the frozen packet. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.5 N V2: dimensional projection

Scientific role. N V2 protects internal units from being misread as laboratory units. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is a dimension identity failure or dependence on arbitrary rescaling. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.6 S/T: unit bridge

Scientific role. S/T controls the external interface. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is an inability to reproduce dimensionless ratios or anchor-dependent quantities transparently. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.7 Y3: particle retest

Scientific role. Y3 shows how to repair an exploratory gap without formula search. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is reappearance of the theta13 gap under the frozen mechanism. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.8 U5/P2: SM scaffold and constants boundary

Scientific role. U5 handles representation compatibility; P2 handles public constants. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is failure of charge, hypercharge, electroweak, or public constant comparison under declared uncertainties. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.9 V2: precision cosmology

Scientific role. V2 is the public likelihood gate. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is poor posterior performance against BAO, SNe, CMB, growth, or matter power when the packet is fixed. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or

external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.10 W2: BBN

Scientific role. W2 exposes the lithium wall. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is failure to reproduce acceptable light elements or inability to address Li7 without damaging other abundances. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.11 X2: CP/EDM/baryogenesis

Scientific role. X2 names the missing EFT and transport layers. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth,

name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is exclusion by EDM bounds or failure to generate baryon asymmetry after physical conversion. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.12 GW2: waveforms

Scientific role. GW2 converts echo language into a null-testable waveform lane. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is absence of statistically meaningful residual structure in public strain data. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.13 Z2: neural/observer

Scientific role. Z2 turns observer concepts into signal statistics. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is failure of pre-registered recursive/fractal measures against controls. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.14 E2: early universe

Scientific role. E2 organizes inflation, transitions, defects, PBHs, relics, and recombination. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is inconsistency with thermal-history or relic constraints under a fixed packet. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.15 QG: quantum geometry

Scientific role. QG tests compatibility with finite spin-foam and geometry audits. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is failure of finite-suite consistency or incompatibility with background-independent constraints. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

H.16 LEG2/TK2: legacy and assembly

Scientific role. LEG2 and TK2 preserve the old story while controlling claim status. Each section should begin with the physical question, not with the result. The reader should know what the module is trying to decide before seeing numbers. A good module section has one job: to connect a triadic role to a calculable or testable object.

Data role. The internal development data for this lane should be divided into four groups. First, carry the frozen G packet and any directly inherited source values. Second, carry the module-specific internal screen or proxy. Third, name the public data required for external comparison. Fourth, name the blocked claim that remains unavailable. This four-part structure prevents the section from blending internal and external status.

Figure role. The best figure for this lane is not decorative. It should show either a dependency chain, a before/after repair, a source-power decomposition, or a boundary dashboard. If the figure cannot be explained in one sentence, it probably belongs in the supplement rather than the main text.

Table role. The table should contain at least one status column. A table without a status column invites overinterpretation. For this lane, the table should state: internal input, computed or carried output, public target, current status, and stronger claim blocked.

Reference role. References should be module-specific. For example, cosmology lanes need Planck, DESI, Pantheon+, CLASS, CAMB, and sampling frameworks. Particle lanes need PDG, CODATA, NIST, BIPM, NuFIT, CKM/PMNS references, and EDM bounds. A generic bibliography is not enough because each lane has its own external standard.

Failure or falsifier. The falsifier is editorial: carrying old claims forward without translation or boundary labels. This statement should remain visible. It is better for the paper to tell reviewers how it can fail than to let them infer that every module is being claimed at the same strength.

Recommended paragraph ending. End the section with a boundary sentence. Do not end with a grand claim. A strong ending looks like this: the lane is internally closed, proxy-bounded, or external-run-ready; the next required work is named; the stronger claim remains blocked until that work is performed.

I Detailed claim-boundary registry

This registry states what the current framework may claim, what it may not claim, and what would be required to move a claim to a stronger status.

Claim topic	Current allowed claim	Stronger claim blocked	Upgrade requirement
Triad	CIF/QV/RFL define three modeling roles	The triad is a proven physical ontology	Formal axiomatization and empirical success across modules
Kernel	A shared recursive skeleton organizes module bases	One equation solves every domain	Domain-specific derivations and tests
Module G	Frozen deterministic packet exists	Packet is physically confirmed	External downstream tests with no retuning
Module R	Source-coupled closure is QV-dominant	Residual audit is empirical validation	Robustness and external tests
Module N	Internal dimensional identities are preserved	Public constants are derived	SI bridge and uncertainty propagation
Cosmology	V2 is likelihood-ready	Hubble tension or dark energy solved	CLASS/CAMB/Cobaya/MontePython runs
BBN	W2 identifies light-element windows and Li7 wall	Lithium problem solved	Public BBN code with reaction network
CP/EDM	X2 defines source windows	Physical EDMs or baryogenesis solved	EFT, Wilson coefficients, transport, bounds
Particle sector	Y3/U5 provide bounded scaffolds	Full Standard Model derived	PDG/NuFIT/CODATA comparison and theory derivation
GW	GW2 defines waveform targets	Echoes detected	GWOSC likelihood and null tests
Neural	Z2 defines signal analysis targets	Consciousness validated	Pre-registered public data analysis
Quantum geometry	QG lanes test finite compatibility	Quantum gravity completed	Formal state-sum proof or broader consistency
Legacy	Older claims are reconciled	All old claims remain valid	Line-by-line claim audit

J Reproducibility

This appendix records the Wolfram/Mathematica input-output packets preserved in the closeout ledger, in the dependency order used by the paper. The purpose is reproducibility, not rhetorical expansion. The packets are shown as internal RFC reproducibility artifacts — they preserve frozen inputs, declared no-retune guards, reported outputs, pass/fail flags, and claim boundaries. They do not by themselves constitute external empirical validation.

The order is: Module G, Module R V3, Module N V2, Modules S/T, QG/QG2/J2, Y3, U/U2-U5, P2, V2, E2, W2, X2, GW2, Z2, LEG2, the combined downstream screen, the dimensionless validation layer, and TK2. Where a lane is carried in the assembly ledger rather than expanded as a separate raw packet, the carried status is reported explicitly.

J.1 Module G - Deterministic triadic closure

```
protocol : MODULE-G
basis : CIF/QV/RFL deterministic triadic closure
target : frozen RFC packet source
score : 1.0000000
final : MODULE-G-FROZEN-PACKET-PASS / NO-RETUNE

FROZEN TRIADIC PACKET
delta : 4.6692000
cycleLength : 60.0000000
triad : 3.0000000
phaseDepthK : 2.0000000
alpha : 0.0256831
nu : 4.208e-3
epsilon : 1.081e-4
lambdaNormalized : 0.4894420
nClosures : 18.0000000
nFullCanonical : 40.0000000

DETERMINISTIC RELATIONS
alpha = Log[delta] / cycleLength
alpha expected : 0.0256831
alpha packet : 0.0256831
alpha abs error : 2.918e-8
alpha pass : YES

nu = phaseDepthK * delta^(-4)
nu expected : 4.208e-3
nu packet : 4.208e-3
nu abs error : 8.434e-9
nu pass : YES

epsilon = alpha * nu
epsilon expected : 1.081e-4
epsilon packet : 1.081e-4
epsilon abs error : 6.245e-10
epsilon pass : YES

NO-RETUNE GUARDRAILS
empirical targets used : NO
parameter search performed : NO
MCMC used : NO
NUTS used : NO
packet frozen : YES
no empirical pass : YES

CLOSURE BREAKDOWN AT n = 18
QV : 0.1449040
CIF : 0.0010014
RFLnorm : 0.7272940
CP : 0.0085615
CollapseRebirth : 0.2805590
Tail : 6.000e-26
closure pass : YES

SYMBOLIC OUTPUTS
symbolicMass : 0.0884241
meanEnergy : 0.0154978
alphaEMsymbolic : 64.5252000
alphaGsymbolic : 0.0078188
LambdaSymbolic : 1.0754200
symbolic pass : YES

CLAIM BOUNDARY
Module G derives the active packet internally from deterministic
CIF-QV-RFL triadic closure. Observational datasets are downstream
validation contexts only.

INTERPRETATION
Module G is the frozen packet source for the active RFC rebuild.
The packet is generated before downstream comparison and must not
be retuned by later validation modules.
```

J.2 Module R V3 - Source-coupled triad closure audit

```

protocol : MODULE-R-V3
basis : Module G frozen packet + source-coupled triad closure audit
target : triad-grouped global closure audit
score : 1.000000
final : MODULE-R-V3-CLOSURE-AUDIT-PASS / NO-RETUNE

FROZEN PACKET INPUT
delta : 4.669200
alpha : 0.025683
nu : 4.208e-3
epsilon : 1.081e-4
lambdaNormalized : 0.489442
nClosure : 18.000000
nFullCanonical : 40.000000
consumes frozen packet : YES
may retune packet : NO

V1 SOURCE-COUPLED AUDIT
raw RFL residual score : 0.733238
source-coupled RFL residual : 0.393570
residual improvement : 0.463244
old collapse-rebirth score : 0.269282
bridge MSE : 0.552744
best lag : -4.000000
best lag correlation : 0.873361
rebirth boundary gap : 0.195142
refined collapse-rebirth : 0.314569
entropy recoil score : 0.041760
observer bifurcation score : 7.481e-6
tail N18 : 6.000e-26
tail N40 : 2.145e-55
old Module R score : 1.002520
refined Module R score : 0.712323
V1 improvement pass : YES

V2 STANDARDIZED RESIDUAL AUDIT
raw standardized residual : 0.990099
source-coupled residual V2 : 0.576064
residual improvement V2 : 0.418175
memory transfer score : 0.146153
Module R score V2 : 0.726401
V2 improvement pass : YES

SOURCE POWER FRACTIONS
QV entropy flow : 0.490033
QV entropy recoil : 0.304474
dark kernel tail : 0.190360
rebirth memory : 0.007827
CIF phase transport : 0.003948
observer branching : 0.002220
CP theta : 0.001137
source power total : 0.999999

TRIAD GROUP POWER
QV core entropy flow/recoil : 0.794507
CIF core phase/CP : 0.005085
RFL core observer/rebirth : 0.010047
QV to RFL dark-kernel bridge : 0.190360

STRICT TRIAD FRACTIONS / NO BRIDGE
QV core : 0.981311
CIF core : 0.006280
RFL core : 0.012409

INTERPRETIVE TRIAD FRACTIONS / WITH DARK BRIDGE
QV inclusive entropy+tail : 0.984867
CIF phase and CP : 0.005085
RFL observer and rebirth : 0.010047

STABILITY CHECKS
memory pass : YES
CP residual : 3.590e-10
CP pass : YES
tail pass : YES
triad power pass : YES
frozen packet pass : YES

CLAIM BOUNDARY
Module R is a source-coupled closure audit of the frozen packet.
It does not create, choose, fit, or retune the packet.

INTERPRETATION
Module R shows that apparent RFL residual structure improves
when treated as source-coupled to QV, CIF, RFL, CP, observer,
dark-kernel, and collapse-rebirth channels. Closure is
overwhelmingly QV-dominant.

```

J.3 Module N V2 - Dimensional projection bridge

```

protocol : MODULE-N-V2
basis : Module G frozen packet + Module R triad closure audit
target : dimension-aware internal RFC projection bridge
score : 1.000000
final : MODULE-N-V2-DIMENSIONAL-PROJECTION-PASS / INTERNAL-UNITS

```

```
FROZEN PACKET INPUT
delta : 4.6692000
cycleLength : 60.0000000
alpha : 0.0256831
phaseDepthK : 2.0000000
nu : 4.208e-3
epsilon : 1.081e-4
lambdaNormalized : 0.4894420
nClosures : 18.0000000
nFullCanonical : 40.0000000
consumes frozen packet : YES
uses Module R audit : YES
may retune packet : NO

MODULE R TRIAD CLOSURE INPUT
residual improvement V2 : 0.4181750
memory transfer score : 0.1461530
best lag : -4.0000000
best lag correlation : 0.8733610
CP residual : 3.590e-10
Module R score V2 : 0.7264010

TRIAD POWER INPUT
QV core power : 0.7945080
dark-kernel bridge : 0.1903600
QV inclusive power : 0.9848680
CIF power : 0.0050847
RFL power : 0.0100470
memory quality : 0.7619930
source closure quality : 0.4181750
triad closure quality : 0.3138250
triad closure pass : YES

CANONICAL BRIDGE
bridge name : QV_Dark_RFL_Memory_CycleBridge
energy factor : 0.2844300
time factor : 1.5409900
cycle time projected : 92.4593000

SYMBOLIC CONSTANTS STABLE ACROSS n = 12, 18, 40
symbolic mass : 0.0872120
mean energy : 0.0154016
alphaEM symbolic : 64.9284000
alphaG symbolic : 0.0076059
Lambda symbolic : 1.0821400

CANONICAL PROJECTED VALUES
symbolic mass projected : 0.0465119
mean energy projected : 0.0043807
alphaEM projected inverse energy : 228.2760000
alphaG projected mass squared : 0.0021634
Lambda projected inverse E cycle : 2.4689300
cycle time projected : 92.4593000
projection pass : YES

IDENTITY CHECKS
alphaG = m^2 residual : 7.520e-9
alphaG = m^2 pass : YES
alphaEM * E = 1 residual : 4.103e-7
alphaEM * E = 1 pass : YES
Lambda * E * T = 1 residual : 5.627e-7
Lambda * E * T = 1 pass : YES
identity pass : YES

CLOSURE
frozen packet pass : YES
Module N pass : YES

CLAIM BOUNDARY
Module N V2 is an internal dimensional projection bridge.
It does not claim final SI laboratory constants.

INTERPRETATION
Module N V2 preserves RFC internal dimensional identities and
prevents raw inverse-energy quantities from being falsely claimed
as physical constants before the S/T bridge and coupling map.
```

J.4 Modules S/T - SI bridge and dimensionless coupling map

```
protocol : MODULE-S-T
basis : Module N projected internal units + frozen G/R/N packet
target : one-anchor SI bridge and triadic dimensionless coupling map
score : 1.000000
final : MODULE-S-T-BRIDGE-PASS / NO-RETUNE

FROZEN PACKET / NO-RETUNE GUARDRAILS
consumes frozen packet : YES
may retune frozen packet : NO
Module G retuned : NO
Module R retuned : NO
Module N retuned : NO
Module S retuned : NO
parameter search performed : NO
empirical targets for deriv. : NO

MODULE S : ONE-ANCHOR SI / LABORATORY BRIDGE
one anchor used : YES
anchor choice : electronRestEnergy
```

```
anchor used as prediction : NO
input RFC mass internal : 0.046355

UNIT BRIDGE
RFC energy unit MeV : 11.023600
RFC energy unit J : 1.766e-12
RFC time unit s : 5.971e-23
RFC length unit m : 1.790e-14
projected cycle time SI s : 5.521e-21
projected cycle length SI m : 1.655e-12

ANCHOR CHECK
projected mass energy MeV : 0.510999
projected mass kg : 9.109e-31
Module S pass : YES

MODULE T : DIMENSIONLESS COUPLING MAP
raw alpha inverse energy : 228.807000
raw alpha claimed physical : NO
canonical map : QVmemRFL

TRIADIC SCREEN
QV inclusive : 0.984868
lambda normalized : 0.489442
Q memory : 0.761993
RFL power : 0.010047
CIF power : 0.005085
screen factor F_T : 0.598472

COUPLING OUTPUT
alpha_T inverse : 136.934528
alpha_T : 0.007303
reference alpha inverse : 137.036000
absolute difference : 0.101472
relative difference : 0.000741
relative difference percent : 0.074046%
within one percent reference : YES
Module T pass : YES

CLAIM BOUNDARY
Module S establishes a one-anchor unit bridge only.
Module T maps an internal inverse-energy quantity into a
dimensionless fine-structure-like coupling through the triadic
QV-memory-RFL screen. The raw value is not claimed as the
physical inverse fine-structure constant.

INTERPRETATION
Module S/T bridges internal RFC projected units into SI-facing
and dimensionless form without retuning the frozen packet.
```

J.5 QG/QG2/J2 - Spin-foam and quantum-geometry suite

```
protocol : QG-QG2-J2
basis : Frozen G/R/N/S/T packet + finite spin-foam quantum-geometry suite
target : finite QG suite closeout and analytic-continuum theorem-readiness ladder
score : 1.000000
final : FINITE-QG-SUITE-PASS / THEOREM-SUPPORT-PASS / FULL-QG-PENDING

FROZEN PACKET / NO-RETUNE
delta : 4.669200
cycleLength : 60.000000
triad : 3.000000
nClosure : 18.000000
nFullCanonical : 40.000000
alpha : 0.025683
nu : 4.208e-3
epsilon : 1.081e-4
lambdaNormalized : 0.489442
consumes frozen packet : YES
may retune packet : NO
empirical targets used : NO
parameter search : NO
no-retune pass : YES

BASELINE MODULE QG FINITE AUDIT
finite amplitude n18 : 1.043940
finite amplitude n40 : 1.043940
relative tail : 4.812e-12
unitarity proxy : 1.000000
refinement proxy : 1.000000
geometry coherence : 0.940535
baseline QG pass : YES

QG2-I FORMAL FINITE-SUITE AUDIT
finite suite checks : 11/11
finite QG score : 0.985348
formal readiness : 7/7
finite suite complete : YES

QG2 SUPPORT CHECKS
domain/gauge checks : 6/6
min domain source : 0.620095
min 4D coherence : 0.999996
domain/gauge pass : YES
RG checks : 15/15
fixed point score : 0.990110
RG beta score : 0.965075
RG/fixed pass : YES
```


KNOWN-MODEL CORRESPONDENCE PROXY

Ponzano-Regge-like : 0.999220
Barrett-Crane-like : 0.998025
EPRL-like : 0.975540
RFC coherence : 0.999996
source score : 0.998923
known-model pass : YES

J2 TRIADIC SPIN-FOAM BIRTH

checks : 25/25
implosion score : 0.854722
kernel implosion : 0.790579
first 4D : 0.240823
first 3D : 0.968467
CP-aware score : 0.994986
birth score : 0.871821
birth pass : YES

J2 COLLAPSE-REBIRTH MEMORY

checks : 15/15
collapse score : 0.054045
memory overlap : 0.958348
lag score : 0.951526
source score : 0.930978
cycle score : 0.993077
memory final score : 0.955425
memory pass : YES

J2 DOWNSTREAM COHERENCE

checks : 16/16
QG lock : 0.946346
dark score : 0.853288
BBN score : 1.000000
CP score : 0.999969
thermal score : 0.930121
CMB score : 0.883555
downstream final score : 0.934329
downstream pass : YES

J2 OBSERVER / RFL REPAIR

checks : 20/20
QG lock D4 : 0.946780
PBH score : 0.978389
psi score : 0.988291
branch score : 0.946341
observer final score : 0.969426
observer pass : YES

J2 LADDER ROBUSTNESS

checks : 12/12
A3 birth : 0.871821
B memory : 0.955425
C bundle : 0.934329
D4 observer : 0.969426
full ladder : 0.931986
robust score : 0.924615
claim boundary : 1.000000
robustness pass : YES

J2-F1 ANALYTIC TAIL SUPPORT

checks : 17/17
continuum score : 0.999845
ladder score : 0.930507
F1 score : 0.975008
F1 pass : YES

J2-FX ANALYTIC-CONTINUUM THEOREM-SUPPORT LADDER

checks : 14/14
F1 tail : 0.993179
F2 refinement : 0.727961
F3 gauge : 1.000000
F4 memory : 0.926952
F5 Regge : 0.986370
FX score : 0.920954
formal ready : YES
written proof : NO
EPRL equivalence : NO
external validation : NO
claim full QG : NO
boundary safe : YES
FX pass : YES

CLOSURE

finite spin-foam suite complete : YES
theorem-support ladder pass : YES
full QG claimed : NO
QG suite pass : YES

INTERPRETATION

QG/QG2/J2 closes the finite spin-foam and quantum-geometry simulation suite and carries an analytic-continuum theorem-support ladder. The written analytic theorem, full EPRL/LQG equivalence, complete transition-amplitude formalism, independent reproduction, peer review, and external validation remain pending.

J.6 Y3 - Frozen particle-sector retest

```
protocol : EXT-Y3-D
basis : Y3-C gate frozen after Y3-A/Y3-B
target : frozen spin-foam particle-sector retest
score : 0.900906
strict : 0.905671
final : FROZEN-PASS
```

```
SUMMARY
protocol : EXT-Y3-D
Y3-A : frozen / YES
Y3-C gate: invK reb / RFLMEM
retune : False / NO
checks : 12/12 / PASS
Y3D score: 0.900906 / PASS
strict : 0.905671 / PASS
final : FROZEN-PASS / FROZEN-PASS
```

```
FREEZE
Y3-A frozen : True / PASS
Y3-B loc : True / PASS
Y3-C cand : True / PASS
gate frozen : True / PASS
new search : False / NO
retune : False / NO
free fit : False / NO
```

```
GATE
gate : invK reb / RFLMEM
kink : 0.184625 / RFL
rebirth : 0.140947 / MEM
gate val : 1.399291 / LOCK
kink-psi : 0.001301 / LOCK
SF score : 0.972150 / PASS
```

```
THETA13
base norm : 0.717672 / LOCK
old err : 28.232840 / GAP
gate : 1.399291 / LOCK
new norm : 1.004231 / PASS
new err : 0.423150 / PASS
improve : 27.809690 / PASS
ratio : 0.985012 / PASS
resolved : True / PASS
```

```
CKM
theta12 % : 0.144070 / PASS
theta23 % : 2.571990 / PASS
theta13 % : 0.423150 / PASS
phase deg : 0.009810 / PASS
sin phase : 0.006740 / PASS
CKM score : 0.910809 / PASS
```

```
SECTORS
quarks : 0.709110 / PASS
CKM : 0.910809 / PASS
PMNS : 0.951898 / PASS
phase : 0.991268 / PASS
unitary : 0.999837 / PASS
collateral : True / PASS
```

```
CLAIMS
theta13 int : True / PASS
full SM : False / PEND
nu mass : False / PEND
external val: False / PEND
Y3-A changed: False / NO
retuned : False / NO
boundary : True / PASS
```

```
FINAL
freeze : 6/6 / PASS
theta13 : 1/1 / PASS
safe : 1/1 / PASS
source : 1/1 / PASS
scores : 2/2 / PASS
claim : 1/1 / PASS
final : FROZEN-PASS / FROZEN-PASS
```

Rule: Y3-D freezes the Y3-C spin-foam-derived RFLMEM gate and retests the particle sector without further search or retuning. PASS means CKM theta13 is internally resolved by the frozen RFL kink/rebirth-memory correction while quarks, CKM core, PMNS, CP phase, and unitarity remain safe. This is not a full Standard Model derivation, neutrino mass-scale derivation, or external independent validation.

J.7 U/U2-U5 - Standard-Model-facing closure ledger

This lane is carried as a closed ledger in the canonical assembly packet. The raw standalone Wolfram packet is not separately expanded in the preserved closeout master, so the carried status is reproduced from TK2.

```
7. U/U2-U5 : Standard-Model-Facing Closure Ledger : YES
closeout pass count : 17/17
all closeouts pass : YES
15. U/U2-U5 - Standard-Model-Facing Closure Ledger : YES
7. U/U2-U5 - Standard-Model-Facing Closure Ledger
```

J.8 P2 - Constants and parameter-table boundary

```
protocol : EXT-P2-H
basis : P2-A through P2-G completed constants / parameter-table chain
target : final constants / parameter-table closure ledger
score : 1.000
final : FINAL-PARAMETER-TABLE-LEDGER-PASS / EXTERNAL-PUBLIC-COMPARISON-BOUNDARY

FROZEN RFC INPUT PACKET
delta : 4.669
cycleLength : 60.000
triad : 3.000
nClosure : 18.000
nFullCanon : 40.000
alpha : 0.026
nu : 0.42e-2
epsilon : 0.11e-3
lambdaNorm : 0.489
targets used : NO

DOWNSTREAM CARRY-FORWARD
QG carry pass : YES
Y carry pass : YES
U carry pass : YES
W2 carry pass : YES
V2 carry pass : YES
X2 carry pass : YES
Z2 carry pass : YES
P2-A carry : YES
P2-B carry : YES
P2-C carry : YES
P2-D carry : YES
P2-E carry : YES
P2-F carry : YES
P2-G carry : YES
all carry pass: YES
no retune : YES

P2 MODULE LEDGER
P2-A : constants / full parameter-table wall diagnosis : YES
P2-B : CODATA / NIST constants public-source audit : YES
P2-C : PDG particle-property / Standard Model table audit : YES
P2-D : unit, dimension, and renormalization-scheme audit : YES
P2-E : RFC proxy-to-public residual map : YES
P2-F : uncertainty / covariance / correlation handoff packet : YES
P2-G : full constants / particle-parameter external-comparison packet : YES
P2 pass count : 7/7
all P2 pass : YES

RFC PARAMETER SCAFFOLD
sin2 thetaW : 0.216
eRFC : 0.303
gRFC : 0.652
gPrime RFC : 0.342
vRFC : 246.000
mW RFC : 80.190
mZ RFC : 90.566
mH carry : 125.100
rho proxy : 1.000
lambdaH : 0.129
muH : 88.459
Vus : 0.216
Vcb : 0.047
Vub : 0.004
J x10^5 : 3.132
phase : 1.047
Yp proxy : 0.249
D/H proxy : 0.26e-4
He3 proxy : 0.10e-4
Li7 proxy : 0.50e-9
Li7 ratio : 3.125
late H0 proxy : 70.173
eta10 proxy : 6.068
sigma8 proxy : 0.808
S8 proxy : 0.808
theta max : 0.11e-3
theta mean : 0.68e-4
CP residual : 0.36e-9
observer divergence : 0.75e-5
fractal coherence : 0.812
geometry coherence : 0.941
memory transfer : 0.146
memory quality : 0.762
symbolic mass projected : 0.046
mean energy projected : 0.004
alphaEM inverse-energy : 228.807
alphaG mass-squared : 0.002
Lambda inverse-energy-cycle : 2.475
RFC scaffold pass : YES

PUBLIC HANDOFF PACKETS
packet 1 : NIST / CODATA source packet : exact SI, measured constants, correlations : YES
packet 2 : PDG / Standard Model packet : particle properties, masses, widths, CKM, PMNS : YES
packet 3 : unit / dimension / scheme packet : comparison guardrails : YES
packet 4 : prototype residual packet : carried anchor residual map : YES
packet 5 : blocked comparison packet : G, exact SI, hadrons, internal units, Li7 : YES
packet 6 : uncertainty / covariance packet : errors, bounds, correlations, posteriors : YES
packet 7 : external comparison packet : full public execution handoff : YES
public packet count : 7
public packet ready : 7
public handoff ready: YES
```

PROXY CLAIMS COMPLETED

claim 1 : internal parameter scaffold closed : YES
claim 2 : CODATA / NIST source layer identified : YES
claim 3 : PDG / SM source layer identified : YES
claim 4 : unit and dimension guardrails closed : YES
claim 5 : renormalization-scheme guardrails closed : YES
claim 6 : prototype residual map closed : YES
claim 7 : blocked comparison map closed : YES
claim 8 : uncertainty / covariance handoff closed : YES
claim 9 : external comparison packet complete : YES
claim 10 : Li7 retained as wall : YES
claim 11 : independent replication required : YES
claim 12 : external public parse required : YES
proxy claim count : 12
proxy claim pass : YES

BLOCKED / UNSOLVED CLAIMS

blocked 1 : G derived : YES
blocked 2 : exact SI constants derived : YES
blocked 3 : elementary charge in coulombs derived : YES
blocked 4 : hadron masses derived : YES
blocked 5 : full quark masses derived : YES
blocked 6 : PMNS / neutrino sector derived : YES
blocked 7 : full Yukawa matrices derived : YES
blocked 8 : decay widths derived : YES
blocked 9 : branching fractions derived : YES
blocked 10 : physical EDMs predicted : YES
blocked 11 : Li7 solved : YES
blocked 12 : full Standard Model derived : YES
blocked 13 : full parameter table externally validated : YES
blocked claim count : 13
blocked claim pass : YES

EXTERNAL EXECUTION BOUNDARY

official sources imported : NO
CODATA parsed : NO
CODATA correlations parsed : NO
PDG parsed : NO
neutrino fit parsed : NO
flavor averages parsed : NO
cosmology public data parsed : NO
BBN public data parsed : NO
EDM bounds parsed : NO
master parameter table built : NO
unit conversion complete : NO
dimension audit complete : NO
scheme audit complete : NO
scale audit complete : NO
uncertainty propagation complete : NO
covariance propagation complete : NO
external residuals computed : NO
z scores computed : NO
chi-square computed : NO
external validation complete : NO
independent replication complete : NO
execution boundary : YES

THEOREM BOUNDARY

external comparison packet ready : YES
theorem boundary ready : YES
external execution not complete : YES
external validation required : YES
independent replication required : YES
theorem-ready not solved : YES
boundary safe : YES

FALSIFIERS

claim full table built fails : YES
claim CODATA parsed fails : YES
claim PDG parsed fails : YES
claim correlations parsed fails : YES
claim residuals computed fails : YES
claim uncertainty done fails : YES
claim covariance done fails : YES
claim external validation fails : YES
claim G derived fails : YES
claim exact SI derived fails : YES
claim hadron masses derived fails : YES
claim full SM derived fails : YES
claim Li7 solved fails : YES
claim physical EDM fails : YES
skip external parse fails : YES
skip boundary fails : NO
retune rejected : YES
falsifier pass : YES

BOUNDARY FLAGS

internal parameter scaffold closed : YES
public constants sources identified : YES
PDG / SM sources identified : YES
unit / dimension guardrails closed : YES
scheme / scale guardrails closed : YES
prototype residual map closed : YES
uncertainty / covariance handoff : YES
external handoff complete : YES
official public data parsed : NO
master table complete : NO
external residuals complete : NO
G derived : NO
exact SI constants derived : NO
hadron masses derived : NO
full Standard Model derived : NO
Li7 solved : NO

physical EDMs predicted : NO
boundary safe : YES

CLOSURE
ledger pass : YES

INTERPRETATION
P2-H closes the constants / full parameter-table refinement lane.
It confirms that RFC now has a complete handoff scaffold for
CODATA/NIST constants, PDG particle properties, unit/dimension
guardrails, renormalization-scheme guardrails, prototype residuals,
blocked comparisons, uncertainty/covariance handling, and public
external comparison.

PASS means theorem-ready and external-public-comparison-ready.

It does not mean official CODATA, PDG, neutrino, flavor, cosmology,
BBN, or EDM public data have been parsed. It does not mean the master
parameter table has been built, residuals have been computed,
uncertainties or correlations have been propagated, G is derived,
exact SI constants are derived, hadron masses are derived, physical
EDMs are predicted, Li7 is solved, or the full Standard Model is derived.

J.9 V2 - Precision cosmology boundary

protocol : EXT-V2-G
basis : V2-A through V2-F completed precision-cosmology chain
target : final precision-cosmology closure ledger
score : 1.000
final : FINAL-PRECISION-COSMO-LEDGER-PASS / LIKELIHOOD-RUN-BOUNDARY

FROZEN RFC INPUT PACKET

delta : 4.669
cycleLength : 60.000
triad : 3.000
nClosure : 18.000
nFullCanon : 40.000
alpha : 0.026
nu : 0.42e-2
epsilon : 0.11e-3
lambdaNorm : 0.489
targets used : NO

DOWNSTREAM CARRY-FORWARD

QG carry pass : YES
Y carry pass : YES
U carry pass : YES
W2 carry pass : YES
all carry pass: YES
no retune : YES

V2 MODULE LEDGER

V2-A : precision-cosmology wall diagnosis : YES
V2-B : BAO / SNe / CMB public-data carry audit : YES
V2-C : expansion-history / Hubble-tension audit : YES
V2-D : dark-energy-tail / w0-wa readiness audit : YES
V2-E : growth / matter-power readiness audit : YES
V2-F : precision-cosmology external-run packet : YES
V2 pass count : 6/6
all V2 pass : YES

INTERNAL COSMOLOGY PROXY

early DM compression : 0.217
late DM compression : 0.038
DM decay ratio : 0.176
DM decay pass : YES
early DE tail : 2.527
late DE tail : 2.484
DE tail ratio : 0.983
DE tail pass : YES
Hubble proxy ratio : 0.959
Hubble proxy pass : YES
internal proxy pass : YES

HUBBLE ANCHOR PACKET

CMB H0 anchor : 67.400
middle H0 anchor : 70.000
high local H0 anchor : 73.000
RFC late H0 proxy : 70.173
RFC bridge fraction : 49.517%
middle residual : 0.173
Hubble window pass : YES

DARK-TAIL WO-WA PACKET

w0 CPL : -1.000
wa CPL : -0.000
near Lambda : YES
w0-wa packet : YES

GROWTH / MATTER-POWER PACKET

sigma8 proxy : 0.808
S8 proxy : 0.808
growth proxy pass : YES
matter power pass : YES
growth/matter pass : YES

PUBLIC DATA TARGETS

data 1 : DESI DR2 BAO : YES

```

data 2 : DESI Ly-alpha BAO : YES
data 3 : Pantheon+ : YES
data 4 : DES-SN5YR : YES
data 5 : Planck PR4 / NPIPE : YES
data 6 : ACT / SPT support : YES
data 7 : matter power / growth data : YES
data 8 : weak lensing data : YES
public data pass : YES

SOLVER / POSTERIOR TARGETS
solver 1 : CLASS background : YES
solver 2 : CLASS perturbations : YES
solver 3 : CAMB background cross-check : YES
solver 4 : CAMB CMB / matter-power cross-check : YES
solver 5 : Cobaya posterior sampling : YES
solver 6 : MontePython posterior cross-check : YES
solver pass : YES

REQUIRED EXTERNAL TASKS
task 1 : BAO likelihood : YES
task 2 : SNe distance-modulus likelihood : YES
task 3 : CMB TT/TE/EE likelihood : YES
task 4 : CMB lensing likelihood : YES
task 5 : CLASS background comparison : YES
task 6 : CLASS perturbation comparison : YES
task 7 : CAMB cross-check : YES
task 8 : matter power P(k) : YES
task 9 : growth fSigma8 : YES
task 10 : weak lensing S8 : YES
task 11 : w0-wa posterior : YES
task 12 : H0 tension audit : YES
task 13 : code-to-code comparison : YES
task 14 : no-retune RFC packet : YES
task 15 : claim-boundary report : YES
task packet pass : YES

REQUIRED EXTERNAL OUTPUTS
output 1 : BAO chi2 : YES
output 2 : SNe chi2 : YES
output 3 : CMB chi2 : YES
output 4 : combined posterior : YES
output 5 : H0 posterior : YES
output 6 : w0-wa posterior : YES
output 7 : Omega_m / sigma8 / S8 : YES
output 8 : P(k) residuals : YES
output 9 : growth residuals : YES
output 10 : model comparison table : YES
output 11 : claim-boundary statement : YES
output packet pass : YES

PROXY CLAIMS COMPLETED
Module V wall diagnosed : YES
public data packet ready : YES
expansion/Hubble window located : YES
dark-tail w0-wa ready : YES
growth/matter-power ready : YES
external-run packet complete : YES
likelihood run required : YES
independent replication required : YES
proxy claim pass : YES

THEOREM BOUNDARY
external packet ready : YES
theorem boundary ready : YES
external execution not complete: YES

FALSIFIERS
claim precision solved fails : YES
claim Hubble solved fails : YES
claim dark energy validated fails: YES
claim growth validated fails : YES
claim likelihood run fails : YES
remove public data fails : YES
remove solver fails : YES
allow retune fails : YES
skip external run fails : YES
falsifier pass : YES

BOUNDARY FLAGS
precision cosmology solved : NO
Hubble tension solved : NO
dark-energy tail validated : NO
growth validated : NO
matter power validated : NO
weak lensing validated : NO
external likelihood complete : NO
Boltzmann comparison complete : NO
posterior completed : NO
full validation complete : NO
external run required : YES
theorem-ready not solved : YES
boundary safe : YES

CLOSURE
ledger pass : YES

INTERPRETATION
V2-G closes the precision-cosmology refinement lane.
It confirms that RFC's internal dark-sector and expansion
proxy can be packaged into BAO, SNe, CMB, Hubble-window,
w0-wa, growth, matter-power, weak-lensing, and posterior
observable packets without retuning.

```

The RFC late-H0 proxy sits near the middle-anchor region rather than claiming to solve the full high-local tension.

PASS means theorem-ready and external-run-ready.
It does not mean precision cosmology is solved, the Hubble tension is solved, dark energy is externally validated, or BAO/SNe/CMB/CLASS/CAMB/posterior runs have been executed.

NEXT

If pass: write precision-cosmology theorem-boundary statement.
Then move to the next weak spot.

J.10 E2 - Early-universe relic and transition boundary

protocol : EXT-E2-H
basis : E2-A through E2-G completed early-universe relic / transition chain
target : final early-universe relic / transition closure ledger
score : 1.000
final : FINAL-EARLY-UNIVERSE-LEDGER-PASS / EXTERNAL-THERMAL-HISTORY-BOUNDARY

FROZEN RFC INPUT PACKET

delta : 4.669
cycleLength : 60.000
triad : 3.000
nClosure : 18.000
nFullCanon : 40.000
alpha : 0.026
nu : 0.42e-2
epsilon : 0.11e-3
lambdaNorm : 0.489
targets used : NO

DOWNSTREAM CARRY-FORWARD

QG carry pass : YES
Y carry pass : YES
U carry pass : YES
W2 carry pass : YES
V2 carry pass : YES
X2 carry pass : YES
Z2 carry pass : YES
P2 carry pass : YES
GW2 carry pass : YES
E2-A carry : YES
E2-B carry : YES
E2-C carry : YES
E2-D carry : YES
E2-E carry : YES
E2-F carry : YES
E2-G carry : YES
all carry pass : YES
no retune : YES

E2 MODULE LEDGER

E2-A : early-universe relic / transition wall diagnosis : YES
E2-B : inflation / reheating readiness audit : YES
E2-C : QCD / electroweak transition audit : YES
E2-D : domain-wall / defect / topological relic packet : YES
E2-E : PBH / early relic abundance packet : YES
E2-F : CMB recombination / thermal-history handoff packet : YES
E2-G : final early-universe external-run packet : YES
E2 pass count : 7/7
all E2 pass : YES

EARLY-UNIVERSE PROXY SCAFFOLD

relative tail : 0.48e-11
geometry coherence : 0.941
memory transfer : 0.146
memory quality : 0.762
collapse-rebirth : 0.315
best lag correlation : 0.873
Yp proxy : 0.249
D/H proxy : 0.26e-4
He3 proxy : 0.10e-4
Li7 proxy : 0.50e-9
Li7 ratio : 3.125
Li7 wall retained : YES
eta10 proxy : 6.068
DM decay ratio : 0.176
DE tail ratio : 0.983
late H0 proxy : 70.173
sigma8 proxy : 0.808
S8 proxy : 0.808
CP source score : 0.545
baryogenesis source : 0.233
washout proxy : 0.533
inflation memory proxy : 0.717
reheating entropy proxy : 0.146
thermal transition proxy : 0.513
relic stability proxy : 0.925
PBH collapse window proxy : 0.055
domain-wall memory proxy : 0.149
CMB recombination carry : 1.513
Neff placeholder proxy : 3.046
e-fold proxy : 60.000
scalar tilt proxy : 0.964
tensor ratio proxy : 0.17e-2
running proxy : -0.56e-3
QCD crossover proxy : 0.925

EW thermal strength proxy : 0.046
domain-wall readiness : 0.140
PBH fraction proxy : 0.34e-4
thermal relic density : 0.140
recombination redshift : 1090.000
CMB spectra readiness : 0.714
energy injection risk : 0.47e-4

PROXY CLAIMS COMPLETED

claim 1 : early-universe wall diagnosed : YES
claim 2 : inflation / reheating packet closed : YES
claim 3 : QCD / EW transition packet closed : YES
claim 4 : domain-wall / defect packet closed : YES
claim 5 : PBH / relic abundance packet closed : YES
claim 6 : CMB recombination packet closed : YES
claim 7 : public source / software packet closed : YES
claim 8 : external task packet closed : YES
claim 9 : external output packet closed : YES
claim 10 : Li7 retained as wall : YES
claim 11 : independent replication required : YES
claim 12 : external thermal-history execution required : YES
proxy claim count : 12
proxy claim pass : YES

BLOCKED / UNSOLVED CLAIMS

blocked 1 : inflation solved : YES
blocked 2 : reheating solved : YES
blocked 3 : primordial perturbation spectrum computed : YES
blocked 4 : tensor spectrum computed : YES
blocked 5 : QCD transition computed : YES
blocked 6 : EW transition computed : YES
blocked 7 : finite-temperature potential computed : YES
blocked 8 : sphaleron / washout dynamics solved : YES
blocked 9 : domain walls physically predicted : YES
blocked 10 : defect network evolved : YES
blocked 11 : wall tension computed : YES
blocked 12 : stochastic GW spectrum computed : YES
blocked 13 : PBH mass function computed : YES
blocked 14 : PBH abundance derived : YES
blocked 15 : PBH constraints scanned : YES
blocked 16 : thermal relic freezeout solved : YES
blocked 17 : thermal relic freezein solved : YES
blocked 18 : nonthermal relic abundance solved : YES
blocked 19 : CMB recombination solved : YES
blocked 20 : CLASS/CAMB spectra generated : YES
blocked 21 : CMB likelihoods run : YES
blocked 22 : external validation completed : YES
blocked claim count : 22
blocked claim pass : YES

EXTERNAL EXECUTION BOUNDARY

inflation solved : NO
slow-roll derived : NO
perturbation spectrum computed: NO
tensor spectrum computed : NO
reheating solved : NO
QCD EOS imported : NO
QCD transition computed : NO
finite-temp EW computed : NO
EW transition computed : NO
sphaleron / washout computed : NO
defect network evolved : NO
wall tension computed : NO
defect GW computed : NO
PBH mass function computed : NO
PBH abundance computed : NO
PBH constraint scan completed : NO
Hawking evaporation computed : NO
thermal freezeout solved : NO
thermal freezein solved : NO
nonthermal relic solved : NO
relic density computed : NO
thermal history integrated : NO
recombination solved : NO
ionization history computed : NO
visibility function computed : NO
sound horizon computed : NO
damping scale computed : NO
CMB spectra computed : NO
CMB lensing computed : NO
CLASS/CAMB run : NO
CMB likelihood run : NO
posterior sampling run : NO
uncertainty propagation done : NO
external validation complete : NO
independent replication : NO
execution boundary : YES

THEOREM BOUNDARY

external packet ready : YES
external run required : YES
independent replication needed : YES
theorem boundary ready : YES
theorem-ready not solved : YES
theorem boundary pass : YES

FALSIFIERS

claim inflation solved fails : YES
claim reheating solved fails : YES
claim perturbation spectrum fails : YES
claim QCD solved fails : YES
claim EW solved fails : YES
claim finite-temp fails : YES


```
claim sphaleron fails : YES
claim defects predicted fails : YES
claim PBH mass function fails : YES
claim PBH abundance fails : YES
claim relic density fails : YES
claim recombination solved fails : YES
claim CMB spectra fails : YES
claim CLASS/CAMB fails : YES
claim likelihood fails : YES
claim external validation fails : YES
retune rejected : YES
falsifier pass : YES
```

BOUNDARY FLAGS

```
internal early proxy closed : YES
public sources identified : YES
external tasks identified : YES
external outputs identified : YES
external handoff complete : YES
official external run complete : NO
inflation solved : NO
reheating solved : NO
QCD / EW transitions solved : NO
domain walls physically predicted : NO
PBHs physically predicted : NO
relic abundance solved : NO
CMB recombination solved : NO
CLASS/CAMB spectra complete : NO
external validation complete : NO
boundary safe : YES
```

CLOSURE

```
ledger pass : YES
```

INTERPRETATION

E2-H closes the early-universe relic / transition refinement lane.
It confirms that RFC now carries a coherent theorem-boundary scaffold for inflation/reheating, QCD and electroweak transitions, topological relics and defects, PBHs, thermal and nonthermal relic abundance, CMB recombination, public software/data targets, external tasks, outputs, falsifiers, and claim boundaries without retuning.

PASS means theorem-ready and external-thermal-history-ready.

It does not mean inflation is solved, reheating is solved, QCD/EW transitions are computed, domain walls are physically predicted, PBH mass functions are computed, relic abundances are solved, recombination is solved, CLASS/CAMB has been run, CMB spectra have been generated, likelihoods have been run, or external validation has been completed.

NEXT

If pass: write early-universe relic / transition theorem-boundary statement.
Then move to the next weak spot or begin assembling the preprint ledger.

J.11 W2 - BBN and lithium-7 boundary

```
protocol : EXT-W2-F
basis : W2-A through W2-E completed Li7 / BBN chain
target : final BBN / Li7 closure ledger
score : 1.000
final : FINAL-BBN-LI7-LEDGER-PASS / EXTERNAL-RUN-BOUNDARY
```

FROZEN RFC INPUT PACKET

```
delta : 4.669
cycleLength : 60.000
triad : 3.000
nClosure : 18.000
nFullCanon : 40.000
alpha : 0.026
nu : 0.42e-2
epsilon : 0.11e-3
lambdaNorm : 0.489
targets used : NO
```

DOWNSTREAM CARRY-FORWARD

```
QG carry pass : YES
Y carry pass : YES
U carry pass : YES
all carry pass: YES
no retune : YES
```

W2 MODULE LEDGER

```
W2-A : BBN / Li7 wall diagnosis : YES
W2-B : reaction-channel carry-forward audit : YES
W2-C2 : Li7 suppression window / no-retune limit : YES
W2-D2 : public BBN code / data-tranche readiness : YES
W2-E : external-run handoff packet / theorem boundary : YES
W2 pass count : 5/5
all W2 pass : YES
```

LIGHT ABUNDANCE STATE

```
Yp proxy : 0.249
Yp error : 0.922%
Yp pass : YES
D/H proxy : 0.26e-4
D/H error : 2.000%
D/H pass : YES
He3 proxy : 0.10e-4
```

```
He3 error : 3.000%
He3 pass : YES
Li7 proxy/network : 0.50e-9
Li7 observed ref : 0.16e-9
Li7 ratio : 3.125
Li7 error : 212.500%
Li7 wall confirmed: YES

LI7 CHANNEL WINDOW
production sens : 0.900
conversion sens : 0.440
destruction sens : 0.500
total sens : 1.840
required suppression : 68.000%
required channel mod : 36.957%
bottleneck located : YES
suppression window : YES

NO-RETUNE LIMIT
small correction max : 0.11e-3
extended frozen max : 0.42e-2
no-retune cannot solve : YES
network calibration : YES

PUBLIC CODE TARGETS
code 1 : PRIMAT
code 2 : AlterBBN
code 3 : PArthENoPE
code 4 : PRyMordial
code 5 : LINX
public code pass : YES

PUBLIC OBSERVABLE TARGETS
observable 1 : Yp helium-4
observable 2 : D/H deuterium
observable 3 : He3/H helium-3
observable 4 : Li7/H lithium-7
observable 5 : eta baryon-photon
observable 6 : neutron lifetime
observable 7 : reaction rates
observable pass : YES

REQUIRED EXTERNAL OUTPUTS
output 1 : Yp
output 2 : D/H
output 3 : He3/H
output 4 : Li7/H
output 5 : reaction sensitivity
output 6 : uncertainty bands
output 7 : code-to-code comparison
output 8 : claim-boundary report
external output pass : YES

REQUIRED NETWORK TASKS
task 1 : temperature-time integration
task 2 : reaction-rate table ingestion
task 3 : weak n-p freeze-out handling
task 4 : baryon density / eta handling
task 5 : light abundance output
task 6 : Li7 channel sensitivity
task 7 : uncertainty propagation
task 8 : multi-code comparison
task 9 : no-retune RFC input packet
task 10 : boundary-safe report
network task pass : YES

PROXY CLAIMS COMPLETED
internal BBN proxy closed : YES
light abundance proxy coherent : YES
Li7 wall localized : YES
Li7 suppression window located : YES
no-retune limit established : YES
public-code handoff ready : YES
external run required : YES
independent replication needed : YES
proxy claim pass : YES

THEOREM BOUNDARY
theorem boundary ready : YES
handoff packet ready : YES
professional network required : YES

FALSIFIERS
claim Li7 solved fails : YES
claim full network run fails : YES
remove public codes fails : YES
remove Li7 observable fails : YES
allow retune fails : YES
skip external run fails : YES
falsifier pass : YES

BOUNDARY FLAGS
full BBN network completed : NO
reaction-rate tables used : NO
temperature-time integration: NO
uncertainty propagation : NO
external code comparison : NO
Li7 solved : NO
Li7 suppression derived : NO
full BBN validation complete : NO
external-run packet complete : YES
theorem-ready not solved : YES
boundary safe : YES
```

```
CLOSURE
ledger pass : YES

INTERPRETATION
W2-F closes the BBN/Li7 refinement lane.
It confirms that RFC carries a coherent light-abundance
proxy for Yp, D/H, and He3/H, while Li7 remains a real
localized wall in the Be7/Li7 channel window.

The no-retune RFC correction scales cannot honestly solve
Li7 by themselves. The lane is therefore closed at an
external-run boundary: public BBN codes, reaction-rate
tables, uncertainty propagation, and independent replication
are required.

PASS means theorem-ready and external-run-ready.
It does not mean Li7 is solved or that a full BBN network
has been executed.

NEXT
If pass: write BBN/Li7 theorem-boundary statement.
Then move to the next weak spot.
```

J.12 X2 - CP, EDM, and baryogenesis boundary

```
protocol : EXT-X2-G
basis : X2-A through X2-F completed CP / EDM / baryogenesis chain
target : final CP / EDM / baryogenesis closure ledger
score : 1.000
final : FINAL-CP-EDM-BARYO-LEDGER-PASS / EXTERNAL-CALCULATION-BOUNDARY

FROZEN RFC INPUT PACKET
delta : 4.669
cycleLength : 60.000
triad : 3.000
nClosure : 18.000
nFullCanon : 40.000
alpha : 0.026
nu : 0.42e-2
epsilon : 0.11e-3
lambdaNorm : 0.489
targets used : NO

DOWNSTREAM CARRY-FORWARD
QG carry pass : YES
Y carry pass : YES
U carry pass : YES
W2 carry pass : YES
V2 carry pass : YES
P2 carry pass : YES
E2 carry pass : YES
all carry pass: YES
no retune : YES

X2 MODULE LEDGER
X2-A : CP / EDM / baryogenesis wall diagnosis : YES
X2-A2 : repaired CP / EDM / baryogenesis wall diagnosis : YES
X2-B : CP-phase / Jarlskog / theta carry audit : YES
X2-C : EDM observable / public-bound readiness audit : YES
X2-D : baryon-eta / baryogenesis source-window audit : YES
X2-E : EFT / Wilson-coefficient readiness audit : YES
X2-F : public-data / external-calculation handoff packet : YES
X2 pass count : 7/7
all X2 pass : YES

CP-PHASE / THETA CARRY
phase : 1.047
J x10-5 : 3.132
theta max : 0.11e-3
theta mean abs : 0.68e-4
CP residual : 0.36e-9
CP source score : 0.545
CP carry pass : YES

EDM BOUND PROXIES
electron EDM proxy : 0.39e-38
electron EDM bound : < 1.1e-29 e cm
electron EDM pass : YES
neutron EDM proxy : 0.34e-30
neutron EDM bound : < 1.8e-26 e cm
neutron EDM pass : YES
EDM suppression N18 : 0.97e-16
EDM suppression N40 : 0.18e-30
fractal suppression : YES
EDM bound pass : YES

BARYON ETA / BARYOGENESIS PROXY
etaB RFC : 0.61e-9
etaB reference : 0.61e-9
etaB error : 0.01475%
baryogenesis source : 0.233
washout proxy : 0.533
source/washout ratio : 0.437
washout survival : 0.304
baryon eta pass : YES

LEGACY PHASE WALL
```

```
legacy CKM phase error before Y2 : 157.128%
legacy PMNS phase error before Y2 : 82.003%
legacy phase wall located : YES
Y3 repair required : YES
Y3 repair downstream : YES
legacy phase boundary pass : YES

EFT / WILSON-COEFFICIENT READINESS
EFT operators identified : YES
Wilson packet ready : YES
EDM observable packet ready : YES
baryogenesis packet ready : YES
CKM/PMNS boundary preserved : YES
public bound sources identified : YES
EFT readiness pass : YES

PUBLIC BOUND / DATA TARGETS
target 1 : ACME electron EDM bound : YES
target 2 : neutron EDM bound : YES
target 3 : hadronic EDM bounds : YES
target 4 : baryon-to-photon eta : YES
target 5 : CKM / Jarlskog public values : YES
target 6 : PMNS phase public values : YES
target 7 : EFT Wilson-coefficient conventions : YES
target 8 : baryogenesis washout literature : YES
public target pass : YES

REQUIRED EXTERNAL TASKS
task 1 : derive EFT operator basis : YES
task 2 : map RFC CP source to Wilson coefficients : YES
task 3 : run electron EDM calculation : YES
task 4 : run neutron EDM calculation : YES
task 5 : propagate hadronic/nuclear uncertainties : YES
task 6 : compute baryogenesis source term : YES
task 7 : compute washout and survival factor : YES
task 8 : compare eta_B to public constraints : YES
task 9 : separate CKM / PMNS phase validation from EDM screen : YES
task 10 : produce no-retune boundary report : YES
external task pass : YES

REQUIRED EXTERNAL OUTPUTS
output 1 : electron EDM prediction : YES
output 2 : neutron EDM prediction : YES
output 3 : hadronic EDM comparison : YES
output 4 : Wilson coefficient table : YES
output 5 : baryogenesis source curve : YES
output 6 : washout/survival curve : YES
output 7 : eta_B residual : YES
output 8 : uncertainty propagation : YES
output 9 : code-to-code comparison : YES
output 10 : claim-boundary statement : YES
required output pass : YES

PROXY CLAIMS COMPLETED
CP wall diagnosed : YES
CP phase carry closed : YES
EDM bound proxy closed : YES
baryon eta proxy closed : YES
legacy phase boundary preserved: YES
EFT packet closed : YES
external packet complete : YES
external calculation required : YES
independent replication required : YES
proxy claim pass : YES

THEOREM BOUNDARY
theorem boundary ready : YES
handoff packet ready : YES
professional EFT calculation req. : YES
theorem-ready not solved : YES
theorem boundary pass : YES

EXTERNAL EXECUTION BOUNDARY
full EFT mapping completed : NO
Wilson running completed : NO
electron EDM derived : NO
neutron EDM derived : NO
hadronic EDM uncertainty done : NO
baryogenesis calculation done : NO
washout calculation done : NO
CKM derived : NO
PMNS derived : NO
external validation complete : NO
independent replication : NO
external execution incomplete : YES

FALSIFIERS
claim full CP/EDM solved fails : YES
claim baryogenesis solved fails: YES
claim physical EDM fails : YES
claim EFT mapped fails : YES
claim Wilson run fails : YES
claim electron EDM fails : YES
claim neutron EDM fails : YES
claim CKM derived fails : YES
claim PMNS derived fails : YES
claim external validation fails: YES
allow retune fails : YES
skip external calculation fails: YES
falsifier pass : YES

BOUNDARY FLAGS
full CP/EDM validation complete : NO
```

```
full baryogenesis proof complete: NO
physical EDM prediction complete: NO
full CKM derivation complete : NO
full PMNS derivation complete : NO
full EFT validation complete : NO
external calculation complete : NO
external calculation required : YES
theorem-ready not solved : YES
boundary safe : YES

CLOSURE
ledger pass : YES

INTERPRETATION
X2-G closes the CP / EDM / baryogenesis refinement lane.
It confirms that the frozen RFC packet carries a stable CP-phase
source, passes electron and neutron EDM bound proxies, preserves
fractal EDM suppression, gives a strong baryon-eta proxy, and packages
the remaining EFT / Wilson / baryogenesis work into an external
calculation boundary without retuning.

PASS means theorem-ready and external-calculation-ready.
It does not mean full EFT mapping, physical EDM derivation, Wilson
running, baryogenesis calculation, CKM/PMNS derivation, or external
validation has been completed.

NEXT
If pass: write CP / EDM / baryogenesis theorem-boundary statement.
Then move to the next weak spot.
```

J.13 GW2 - Gravitational-wave and ringdown boundary

```
protocol : EXT-GW2-G
basis : GW2-A through GW2-F completed gravitational-wave / ringdown / echo chain
target : final gravitational-wave / ringdown / echo closure ledger
score : 1.000
final : FINAL-GW-LEDGER-PASS / EXTERNAL-WAVEFORM-BOUNDARY
```

```
FROZEN RFC INPUT PACKET
delta : 4.669
cycleLength : 60.000
triad : 3.000
nClosure : 18.000
nFullCanon : 40.000
alpha : 0.026
nu : 0.42e-2
epsilon : 0.11e-3
lambdaNorm : 0.489
targets used : NO
```

```
DOWNSTREAM CARRY-FORWARD
QG carry pass : YES
Y carry pass : YES
U carry pass : YES
W2 carry pass : YES
V2 carry pass : YES
X2 carry pass : YES
P2 carry pass : YES
E2 carry pass : YES
all carry pass: YES
no retune : YES
```

```
GW2 MODULE LEDGER
GW2-A : gravitational-wave / ringdown / echo wall diagnosis : YES
GW2-B : ringdown observable / quasinormal-mode packet : YES
GW2-C : echo-delay / late-time residual window : YES
GW2-D : public GW data / software source audit : YES
GW2-E : waveform / likelihood / Bayesian boundary : YES
GW2-F : external-run handoff packet : YES
GW2 pass count : 6/6
all GW2 pass : YES
```

```
GEOMETRY / MEMORY CARRY
relative tail : 0.48e-11
geometry coherence : 0.941
spin-foam initial weight : 0.875
spin-foam final weight : 0.977
spin-foam final/initial : 1.116
QG lock score : 0.946
observer lock score : 0.969
robust ladder score : 0.925
geometry carry pass : YES
memory transfer score : 0.146
memory quality : 0.762
best lag : -4.000
best lag correlation : 0.873
collapse-rebirth score : 0.315
rebirth boundary gap : 0.195
memory carry pass : YES
```

```
RINGDOWN / ECHO PROXY
echo delay proxy : 20.000
echo lower window : 18.000
echo upper window : 22.000
echo window pass : YES
ringdown damping proxy : 0.941
late residual proxy : 0.146
phase coherence proxy : 0.985
```

```
recursive echo persistence : 0.925
QNM coherence proxy : 0.934
waveform stability proxy : 0.947
ringdown observable pass : YES
echo residual pass : YES

PUBLIC GW DATA TARGETS
target 1 : GWOSC public strain data : YES
target 2 : LIGO/Virgo/KAGRA event catalogs : YES
target 3 : ringdown event subset : YES
target 4 : black-hole merger waveform releases : YES
target 5 : noise PSD / calibration products : YES
target 6 : posterior samples where public : YES
target 7 : injection / recovery comparison target : YES
public GW pass : YES

SOFTWARE TARGETS
software 1 : Bilby : YES
software 2 : PyCBC : YES
software 3 : LALSuite : YES
software 4 : GWpy : YES
software 5 : ringdown / QNM fitting package : YES
software 6 : Bayesian evidence / model-comparison tool : YES
software pass : YES

REQUIRED EXTERNAL WAVEFORM TASKS
task 1 : download public strain data : YES
task 2 : apply event-quality vetoes : YES
task 3 : estimate PSD and whitening : YES
task 4 : fit GR baseline ringdown : YES
task 5 : fit RFC recursive residual / echo template : YES
task 6 : compare QNM residuals : YES
task 7 : run injection recovery : YES
task 8 : compute Bayes factor : YES
task 9 : estimate posterior for echo delay : YES
task 10 : perform code-to-code comparison : YES
task 11 : preserve no-retune RFC packet : YES
task 12 : produce claim-boundary report : YES
external task pass : YES

REQUIRED EXTERNAL OUTPUTS
output 1 : GR baseline waveform residuals : YES
output 2 : RFC residual / echo waveform residuals : YES
output 3 : posterior echo-delay window : YES
output 4 : Bayes factor / evidence comparison : YES
output 5 : QNM residual table : YES
output 6 : injection-recovery table : YES
output 7 : event-by-event robustness table : YES
output 8 : noise-systematics audit : YES
output 9 : code-to-code comparison : YES
output 10 : claim-boundary statement : YES
required output pass : YES

PROXY CLAIMS COMPLETED
GW wall diagnosed : YES
ringdown observable packet : YES
echo window located : YES
public data packet ready : YES
software packet ready : YES
waveform likelihood boundary : YES
external packet complete : YES
external waveform run required : YES
independent replication required : YES
proxy claim pass : YES

THEOREM BOUNDARY
theorem boundary ready : YES
handoff packet ready : YES
professional waveform analysis req. : YES
theorem-ready not solved : YES
theorem boundary pass : YES

EXTERNAL EXECUTION BOUNDARY
public strain downloaded : NO
event selection completed : NO
PSD / whitening completed : NO
GR ringdown fit completed : NO
RFC waveform fit completed : NO
QNM residuals computed : NO
echo posterior computed : NO
Bayes factor computed : NO
injection recovery completed : NO
noise systematics completed : NO
code-to-code comparison : NO
external validation complete : NO
independent replication : NO
external execution incomplete : YES

FALSIFIERS
claim GW validated fails : YES
claim echo detected fails : YES
claim waveform derived fails : YES
claim GR replacement fails : YES
claim catalog validated fails : YES
claim Bayes factor done fails : YES
claim external run done fails : YES
claim strain parsed fails : YES
allow retune fails : YES
skip external run fails : YES
falsifier pass : YES

BOUNDARY FLAGS
full GW validation complete : NO
```

```
physical echo detection claimed : NO
full waveform model derived : NO
GR replacement claimed : NO
event catalog validation complete: NO
Bayesian comparison complete : NO
external waveform run complete : NO
external waveform run required : YES
theorem-ready not solved : YES
boundary safe : YES

CLOSURE
ledger pass : YES

INTERPRETATION
GW2-G closes the gravitational-wave / ringdown / echo refinement lane.
It confirms that the frozen RFC packet carries a coherent ringdown
and recursive residual/echo proxy scaffold, supported by geometry,
spin-foam coherence, collapse-rebirth memory, and no-retune guardrails.

PASS means theorem-ready and external-waveform-run-ready.
It does not mean a physical echo detection has been made, public strain
data have been parsed, full waveform inference has been run, Bayes
factors have been computed, GR has been replaced, or external validation
has been completed.
```

J.14 Z2 - Observer, neural, and EEG boundary

```
protocol : EXT-Z2-G
basis : Z2-A through Z2-F completed neural / EEG chain
target : final neural / EEG closure ledger
score : 1.000
final : FINAL-NEURAL-EEG-LEDGER-PASS / EXTERNAL-ANALYSIS-BOUNDARY
```

```
FROZEN RFC INPUT PACKET
delta : 4.669
cycleLength : 60.000
triad : 3.000
nClosure : 18.000
nFullCanon : 40.000
alpha : 0.026
nu : 0.42e-2
epsilon : 0.11e-3
lambdaNorm : 0.489
targets used : NO
```

```
DOWNSTREAM CARRY-FORWARD
QG carry pass : YES
Y carry pass : YES
U carry pass : YES
W2 carry pass : YES
V2 carry pass : YES
X2 carry pass : YES
P2 carry pass : YES
E2 carry pass : YES
GW2 carry pass: YES
all carry pass: YES
no retune : YES
```

```
Z2 MODULE LEDGER
Z2-A : neural / EEG wall diagnosis : YES
Z2-B : public EEG / MEG / iEEG data-target audit : YES
Z2-C : EEG preprocessing / signal-readiness audit : YES
Z2-D : neural avalanche / criticality observable audit : YES
Z2-E : fractal / recursive-signature audit : YES
Z2-F : public-data / external-analysis handoff packet : YES
Z2 pass count : 6/6
all Z2 pass : YES
```

```
OBSERVER / BRANCHING MODEL SCREEN
observer divergence : 0.75e-5
observer target : < 1e-5
observer pass : YES
branch final/max : 0.060
branch target : < 0.1
branch pass : YES
decoherence proxy : 0.57e-7
decoherence target : < 1e-6
decoherence pass : YES
memory correlation : 0.873
memory target : > 0.8
memory pass : YES
observer/branching pass : YES
```

```
NEURAL / EEG TARGET SIGNATURES
neural fractal D : 2.489
neural fractal D target : 2.450
neural fractal D error : 1.60980%
avalanche exponent : 1.489
avalanche target : 1.500
avalanche error : 0.70400%
spectral slope : 1.974
spectral slope target : 2.000
spectral slope error : 1.28400%
target signature pass : YES
```

```
RECURSIVE / FRACTAL SIGNATURE SCAFFOLD
fractal coherence proxy : 0.812
geometry coherence mean : 0.941
```

```
memory transfer score : 0.146
memory quality : 0.762
best lag : -4.000
best lag correlation : 0.873
observer lock score : 0.969
branch score : 0.946
psi score : 0.988
recursive signature pass : YES

PUBLIC DATA TARGETS
target 1 : OpenNeuro EEG datasets : YES
target 2 : PhysioNet EEG Motor Movement / Imagery : YES
target 3 : CHB-MIT EEG / seizure benchmark : YES
target 4 : HCP / MEG-style public target : YES
target 5 : iEEG public benchmark target : YES
target 6 : neural avalanche dataset target : YES
target 7 : sleep / resting-state EEG target : YES
target 8 : clinical EEG comparison target : YES
public data pass : YES

PREPROCESSING / SIGNAL-READINESS TASKS
preprocess 1 : raw data import : YES
preprocess 2 : channel montage harmonization : YES
preprocess 3 : notch filtering : YES
preprocess 4 : bandpass filtering : YES
preprocess 5 : artifact rejection / ICA : YES
preprocess 6 : epoching / windowing : YES
preprocess 7 : subject-level normalization : YES
preprocess 8 : cross-dataset split : YES
preprocess 9 : no-retune RFC packet preservation : YES
preprocess 10 : claim-boundary report : YES
preprocessing pass : YES

REQUIRED EXTERNAL ANALYSIS TASKS
task 1 : fractal dimension extraction : YES
task 2 : avalanche exponent extraction : YES
task 3 : spectral slope extraction : YES
task 4 : criticality / branching-ratio audit : YES
task 5 : memory-correlation test : YES
task 6 : observer-branching proxy comparison : YES
task 7 : cross-subject robustness : YES
task 8 : cross-dataset robustness : YES
task 9 : null-model comparison : YES
task 10 : surrogate-data comparison : YES
task 11 : uncertainty propagation : YES
task 12 : independent replication packet : YES
analysis task pass : YES

REQUIRED EXTERNAL OUTPUTS
output 1 : fractal dimension table : YES
output 2 : avalanche exponent table : YES
output 3 : spectral slope table : YES
output 4 : criticality / branching table : YES
output 5 : memory-correlation table : YES
output 6 : subject-level residuals : YES
output 7 : dataset-level residuals : YES
output 8 : surrogate comparison table : YES
output 9 : uncertainty / confidence intervals : YES
output 10 : external analysis report : YES
output 11 : claim-boundary statement : YES
required output pass : YES

PROXY CLAIMS COMPLETED
neural wall diagnosed : YES
observer/branching closed : YES
target signature packet closed : YES
recursive signature closed : YES
public data packet ready : YES
preprocessing packet ready : YES
external analysis packet : YES
external neural data required : YES
independent replication required : YES
proxy claim pass : YES

THEOREM BOUNDARY
theorem boundary ready : YES
handoff packet ready : YES
professional neural analysis req. : YES
theorem-ready not solved : YES
theorem boundary pass : YES

EXTERNAL EXECUTION BOUNDARY
real EEG data imported : NO
real MEG data imported : NO
real iEEG data imported : NO
preprocessing executed : NO
fractal dimension measured : NO
avalanche exponent measured : NO
spectral slope measured : NO
criticality analysis executed : NO
memory correlation measured : NO
subject-level statistics : NO
dataset-level statistics : NO
surrogate analysis executed : NO
uncertainty propagation complete : NO
external validation complete : NO
independent replication : NO
external execution incomplete : YES

FALSIFIERS
claim consciousness proven fails : YES
claim observer branching fails : YES
claim many-worlds proven fails : YES
```



```
claim EEG validation fails : YES
claim real data imported fails : YES
claim preprocessing done fails : YES
claim fractal measured fails : YES
claim avalanche measured fails : YES
claim spectral measured fails : YES
claim external validation fails : YES
allow retune fails : YES
skip external data fails : YES
falsifier pass : YES

BOUNDARY FLAGS
consciousness experimentally proven : NO
observer branching experimentally proven : NO
many-worlds experimentally proven : NO
neural / EEG validation complete : NO
real dataset validation complete : NO
clinical prediction claimed : NO
external analysis complete : NO
external neural data required : YES
theorem-ready not solved : YES
boundary safe : YES

CLOSURE
ledger pass : YES

INTERPRETATION
Z2-G closes the neural / EEG refinement lane.
It confirms that RFC carries a coherent observer/branching model
screen, neural fractal target signature, avalanche target signature,
spectral-slope target signature, recursive memory scaffold, public
data-target packet, preprocessing packet, and external-analysis packet
without retuning.

PASS means theorem-ready and external-analysis-ready.
It does not mean consciousness is experimentally proven, observer
branching is experimentally proven, many-worlds physics is proven,
real EEG/MEG/iEEG data have been imported, preprocessing has been
run, neural signatures have been measured on external datasets,
or external validation has been completed.

NEXT
If pass: write neural / EEG theorem-boundary statement.
Then move to legacy reconciliation.
```

J.15 LEG2 - Legacy reconciliation ledger

```
protocol : EXT-LEG2-G
basis : LEG2-A through LEG2-F completed legacy reconciliation chain
target : final legacy reconciliation closure ledger
score : 1.000
final : FINAL-LEGACY-RECONCILIATION-LEDGER-PASS / SOURCE-BACKED-RECONCILIATION-BOUNDARY

FROZEN RFC INPUT PACKET
delta : 4.669
cycleLength : 60.000
triad : 3.000
nClosure : 18.000
nFullCanon : 40.000
alpha : 0.026
nu : 0.42e-2
epsilon : 0.11e-3
lambdaNorm : 0.489
targets used : NO

DOWNSTREAM CARRY-FORWARD
QG carry pass : YES
Y3 carry pass : YES
U carry pass : YES
P2 carry pass : YES
V2 carry pass : YES
E2 carry pass : YES
W2 carry pass : YES
X2 carry pass : YES
GW2 carry pass : YES
Z2 carry pass : YES
all carry pass : YES
no retune : YES

LEG2 MODULE LEDGER
LEG2-A : legacy source inventory / preprint map : YES
LEG2-B : deprecated MCMC / old Module G reconciliation : YES
LEG2-C : legacy module A-Q downstream-status audit : YES
LEG2-D : old claims reclassification / boundary audit : YES
LEG2-E : new canonical module mapping audit : YES
LEG2-F : source-backed reconciliation packet : YES
LEG2 pass count : 6/6
all LEG2 pass : YES

SOURCE INVENTORY
source 1 : Presentation 24 : old simulation validation / MCMC / echo / BBN-halo fit context : YES
source 2 : Presentation 25 : intermediate RFC module-development context : YES
source 3 : Presentation 26 : triad-to-module mapping / empirical anchoring context : YES
source 4 : Presentation 27 : canonical triad-first deterministic rebuild : YES
source 5 : The rebuild.docx : new preprint scaffold and deterministic replacement logic : YES
source 6 : PDE_Kernels.txt : kernel and module status documentation : YES
source 7 : CLAIM_BOUNDARIES.md : active claim-boundary and legacy status table : YES
source 8 : Appendix.txt : appendix protocol and deprecated-module statements : YES
```

```
source 9 : SimulationConfigs.json : repository configuration and module metadata : YES
source 10 : Module_G_R_N_S_T_FrozenPacket.json : active frozen packet source : YES
source count : 10
source pass : YES

DEPRECATED ACTIVE-DERIVATION ITEMS
old MCMC Module G deprecated : YES
old NUTS deprecated : YES
old best-fit epsilon deprecated : YES
old observational target fitting dep.: YES
old external targets upstream dep. : YES
old symbolic-fractal R active dep. : YES
old raw constants physical dep. : YES
deprecated pass : YES

ACTIVE REPLACEMENT MAP
old MCMC Module G -> Deterministic Triadic Closure : YES
old symbolic-fractal Module R -> Source-Coupled Triad Closure Audit : YES
old raw constants bridge -> Dimensional Projection Bridge + SI Bridge / Dimensionless Coupling Map : YES
old spin-foam coherence proxy -> Finite Spin-Foam / QG Suite : YES
old particle-sector exploratory claim -> Frozen Particle-Sector Retest : YES
old constants / SM-facing scaffold -> Standard-Model-Facing Closure Ledger + P2 : YES
old BBN / cosmology / CP / EEG screens -> P2/V2/E2/W2/X2/GW2/Z2 theorem-boundary lanes : YES
active mapping pass : YES

LEGACY MODULE STATUS MAP
module A : historical / downstream development context : YES
module B : historical / downstream development context : YES
module C : historical / downstream development context : YES
module D : historical / downstream development context : YES
module E : historical / downstream development context : YES
module F : retained observer-field downstream projection : YES
module G_legacy : deprecated MCMC / empirical assimilation only : YES
module H : geometry coherence context carried into QG : YES
module I : historical / downstream development context : YES
module J : historical / downstream development context : YES
module K : collapse-rebirth memory context retained : YES
module L : historical / downstream development context : YES
module M : historical symbolic-constant context only : YES
module N_legacy : deprecated / replaced by N V2 bridge : YES
module O : historical / downstream development context : YES
module P : baryogenesis / dark-sector downstream proxy context : YES
module Q : scalar-induced observer decoherence downstream proxy : YES
module R_legacy : deprecated symbolic-fractal unification as active derivation : YES
legacy module count : 18
legacy module pass : YES

OLD CLAIM RECLASSIFICATION
old claim 1 : parameter sweep selected epsilon -> historical only; replaced by deterministic epsilon : YES
old claim 2 : epsilon = 3.2e-4 best-fit packet -> deprecated; active epsilon = 0.000108071 : YES
old claim 3 : old BBN / halo residual fit -> reclassified as downstream/proxy context : YES
old claim 4 : old constants derivation -> replaced by N V2, S, T, U, and P2 boundary : YES
old claim 5 : old particle-sector validation -> replaced by Y2 exploratory then Y3 frozen retest : YES
old claim 6 : old observer / consciousness claims -> reframed as observer/neural model-screen boundary : YES
old claim 7 : old spin-foam coherence proxy -> expanded into QG/QG2/J2 finite suite boundary : YES
old claim 8 : old echo comparison -> reframed into GW2 waveform-run boundary : YES
old claim 9 : old CP / asymmetry claims -> reframed into X2 EFT/external-calculation boundary : YES
reclassification pass : YES

CANONICAL CURRENT MODULE ORDER
1. G
2. R
3. N
4. S/T
5. QG/QG2/J2
6. Y3
7. U/U2-U5
8. P2
9. V2
10. E2
11. W2
12. X2
13. GW2
14. Z2
15. LEG2
canonical order pass : YES

ACTIVE BOUNDARY FLAGS
old Module G active : NO
old MCMC active : NO
old NUTS active : NO
old empirical target fitting active: NO
old best-fit epsilon active : NO
legacy modules retune packet : NO
legacy modules override packet : NO
legacy claims override boundaries : NO
Presentation 24 overrides P27 : NO
Presentation 25 overrides P27 : NO
Presentation 26 overrides P27 : NO
source conflict unresolved : NO
boundary safe : YES

EXTERNAL / PROOF BOUNDARY
legacy reconciled internally : YES
source-backed reconciliation : YES
preprint-ready reconciliation : YES
external validation complete : NO
peer review complete : NO
independent replication complete : NO
full laboratory theory claimed : NO
external boundary pass : YES

FALSIFIERS
```

```
claim old MCMC still active fails : YES
claim old G still active fails : YES
claim old best-fit epsilon active fails : YES
claim legacy retunes packet fails : YES
claim legacy overrides packet fails : YES
claim old preprints override new fails : YES
claim source conflict unresolved fails : YES
claim external validation complete fails: YES
claim full lab theory fails : YES
falsifier pass : YES
```

```
CLOSURE
ledger pass : YES
```

```
INTERPRETATION
LEG2-G closes the legacy reconciliation lane.
It confirms that the older RFC development record is retained as
source-backed history and downstream context, while the active
preprint architecture is the triad-first deterministic rebuild.
```

Old MCMC/NUTS Module G, old best-fit epsilon, old upstream empirical target fitting, and older over-broad validation language are deprecated as active derivation. They remain useful only as development history.

PASS means the legacy record is reconciled into the current module

J.16 Combined downstream physical-projection screen

The rendered status for this screen is carried by the TK2 assembly packet. The preserved Wolfram code/input packet is reproduced here because it contains the module inputs, checks, and output-string construction.

order and claim-boundary system.

It does not mean old claims override the current frozen packet, external validation is complete, peer review is complete, independent replication is complete, or RFC is a fully validated laboratory theory.

```
NEXT
If pass: move to TK2 / canonical preprint assembly, or place this
ledger in the appendix as the final legacy reconciliation packet.
```

```
(* COMBINED DOWNSTREAM PHYSICAL-PROJECTION SCREEN *)
ClearAll["Global`*"];
```

```
protocol = "COMBINED-PHYSICAL-PROJECTION";
basis = "Frozen Module G packet + Module R closure + Module N V2 projection";
target = "downstream physical-projection screen across retained RFC domains";
```

```
(* frozen Module G packet *)
delta = 4.6692;
cycleLength = 60.;
triad = 3.;
nClosure = 18.;
nFullCanonical = 40.;
alpha = 0.0256831;
phaseDepthK = 2.;
nu = 0.00420784;
epsilon = 0.000108071;
lambdaNormalized = 0.489442;
empiricalTargetsUsed = False;
```

```
(* no-retune guardrails *)
usesFrozenModuleG = True;
usesModuleRClosure = True;
usesModuleNV2Projection = True;
physicalComparisonPerformed = False;
mayRetuneFrozenPacket = False;
parameterSearchPerformed = False;
```

```
(* Module N projected combined-screen values *)
symbolicMass = 0.0869182;
meanEnergy = 0.0153658;
alphaEMsymbolic = 65.0795;
alphaGsymbolic = 0.00755478;
LambdaSymbolic = 1.08466;
```

```
symbolicMassProjected = 0.0463552;
meanEnergyProjected = 0.0043705;
alphaEMProjectedInverseEnergy = 228.807;
alphaGProjectedMassSquared = 0.00214881;
LambdaProjectedInverseEnergyCycle = 2.47467;
```

```
massSquaredIdentityResidual =
  Abs[alphaGProjectedMassSquared - symbolicMassProjected^2];
```

```
inverseEnergyIdentityResidual =
  Abs[alphaEMProjectedInverseEnergy*meanEnergyProjected - 1];
```

```
lambdaCycleIdentityResidual =
  Abs[LambdaProjectedInverseEnergyCycle*meanEnergyProjected*92.4594 - 1];
```

```
moduleNIdentityPass =
  massSquaredIdentityResidual < 10^-6 &&
  inverseEnergyIdentityResidual < 10^-5 &&
  lambdaCycleIdentityResidual < 10^-5;
```

```

(* Module A: dark-sector cosmology projection *)
earlyDarkMatterCompression = 0.216534;
lateDarkMatterCompression = 0.0381948;
darkMatterDecayRatio = 0.176392;

earlyDarkEnergyTail = 2.52655;
lateDarkEnergyTail = 2.48382;
darkEnergyTailRatio = 0.983088;

earlyHubbleProxy = 1.65623;
lateHubbleProxy = 1.58809;
hubbleProxyRatio = 0.958858;

darkSectorPass =
  lateDarkMatterCompression < earlyDarkMatterCompression &&
  darkMatterDecayRatio < 0.25 &&
  darkEnergyTailRatio > 0.95 &&
  darkEnergyTailRatio < 1.02 &&
  hubbleProxyRatio > 0.9 &&
  hubbleProxyRatio < 1.05;

(* Module C: BBN primordial proxy *)
initialYp = 0.246;
finalYp = 0.249278;
deltaYp = 0.00327757;
gammaZetaInitial = epsilon;
gammaZetaFinal = 0.0000231454;

bbnProxyPass =
  finalYp > initialYp &&
  deltaYp > 0 &&
  finalYp > 0.24 &&
  finalYp < 0.26 &&
  gammaZetaFinal < gammaZetaInitial;

(* Module B: CP / EDM proxy *)
thetaMax = 0.000108055;
thetaMeanAbs = 0.000068068;
cpResidual = 3.59079*10^-10;
fractalEDMSuppressionN18 = 9.71271*10^-17;
fractalEDMSuppressionN40 = 1.83655*10^-31;

cpEDMPass =
  thetaMax > 0 &&
  thetaMeanAbs > 0 &&
  cpResidual < 10^-8 &&
  fractalEDMSuppressionN18 < 10^-12 &&
  fractalEDMSuppressionN40 < fractalEDMSuppressionN18;

(* Module F/J: observer branching *)
observerDivergence = 7.49536*10^-6;
branchSeparationMax = 0.0000232443;
branchSeparationFinal = 1.40621*10^-6;
decoherenceProxy = 5.73826*10^-8;

observerBranchingPass =
  observerDivergence < 10^-4 &&
  branchSeparationFinal < branchSeparationMax &&
  decoherenceProxy < 10^-6;

(* Module K: collapse-rebirth memory *)
rebirthInitial = 0.148775;
rebirthFinal = -0.0463672;
rebirthBoundaryGap = 0.195142;
bestLag = -5;
bestLagCorrelation = 0.905876;
memoryTransferScore = 0.113638;

collapseRebirthPass =
  rebirthBoundaryGap > 0 &&
  Abs[bestLag] >= 1 &&
  bestLagCorrelation > 0.8 &&
  memoryTransferScore > 0;

(* Module E/H: fractal geometry *)
normalizedPsiRMS = 1.;
fractalCoherenceProxy = 0.812251;
spinFoamWeightInitial = 0.875122;
spinFoamWeightFinal = 0.976745;
geometryCoherenceMean = 0.940535;

geometryPass =
  normalizedPsiRMS == 1. &&
  fractalCoherenceProxy > 0.8 &&
  spinFoamWeightFinal > spinFoamWeightInitial &&
  geometryCoherenceMean > 0.9;

noRetunePass =
  usesFrozenModuleG &&
  usesModuleRClosure &&
  usesModuleNV2Projection &&
  ! empiricalTargetsUsed &&
  ! physicalComparisonPerformed &&
  ! mayRetuneFrozenPacket &&
  ! parameterSearchPerformed;

combinedProjectionPass =
  noRetunePass &&
  moduleIdentityPass &&
  darkSectorPass &&
  bbnProxyPass &&
  cpEDMPass &&

```

```

observerBranchingPass &&
collapseRebirthPass &&
geometryPass;

coreChecks = {
  noRetunePass,
  moduleNIdentityPass,
  darkSectorPass,
  bbnProxyPass,
  cpEDMPass,
  observerBranchingPass,
  collapseRebirthPass,
  geometryPass
};

score = N[Count[coreChecks, True]/Length[coreChecks]];

final =
  If[combinedProjectionPass,
    "COMBINED-PHYSICAL-PROJECTION-PASS / INTERNAL-PROJECTED-UNITS",
    "CHECK / WALL"];

yn[x_] := If[TrueQ[x], "YES", "NO"];
nf[x_] := ToString[NumberForm[N[x], {10, 6}]];
sci[x_] := Module[{xx, me},
  xx = N[x];
  If[Abs[xx] < 10^-14,
    "0",
    me = MantissaExponent[xx];
    ToString[NumberForm[me[[1]], {6, 3}]] <> "e" <> ToString[me[[2]]]]
];

out = StringRiffle[
{
  "protocol : " <> protocol,
  "basis : " <> basis,
  "target : " <> target,
  "score : " <> nf[score],
  "final : " <> final,
  "",
  "FROZEN PACKET / NO-RETUNE",
  "delta : " <> nf[delta],
  "cycleLength : " <> nf[cycleLength],
  "triad : " <> nf[triad],
  "nClosure : " <> nf[nClosure],
  "nFullCanonical : " <> nf[nFullCanonical],
  "alpha : " <> nf[alpha],
  "nu : " <> sci[nu],
  "epsilon : " <> sci[epsilon],
  "lambdaNormalized : " <> nf[lambdaNormalized],
  "uses frozen G : " <> yn[usesFrozenModuleG],
  "uses Module R : " <> yn[usesModuleRClosure],
  "uses Module N V2 : " <> yn[usesModuleNV2Projection],
  "empirical targets : " <> yn[empiricalTargetsUsed],
  "physical compare : " <> yn[physicalComparisonPerformed],
  "retune allowed : " <> yn[mayRetuneFrozenPacket],
  "parameter search : " <> yn[parameterSearchPerformed],
  "no-retune pass : " <> yn[noRetunePass],
  "",
  "MODULE N PROJECTED COMBINED SCREEN",
  "symbolic mass : " <> nf[symbolicMass],
  "mean energy : " <> nf[meanEnergy],
  "alphaEM symbolic : " <> nf[alphaEMsymbolic],
  "alphaG symbolic : " <> nf[alphaGsymbolic],
  "Lambda symbolic : " <> nf[LambdaSymbolic],
  "symbolic mass projected : " <> nf[symbolicMassProjected],
  "mean energy projected : " <> nf[meanEnergyProjected],
  "alphaEM inverse-energy : " <> nf[alphaEMProjectedInverseEnergy],
  "alphaG mass-squared : " <> nf[alphaGProjectedMassSquared],
  "Lambda inverse-energy-cycle : " <> nf[LambdaProjectedInverseEnergyCycle],
  "mass-squared residual : " <> sci[massSquaredIdentityResidual],
  "inverse-energy residual : " <> sci[inverseEnergyIdentityResidual],
  "lambda-cycle residual : " <> sci[lambdaCycleIdentityResidual],
  "Module N identity pass : " <> yn[moduleNIdentityPass],
  "",
  "MODULE A : DARK-SECTOR COSMOLOGY PROJECTION",
  "early dark-matter compression : " <> nf[earlyDarkMatterCompression],
  "late dark-matter compression : " <> nf[lateDarkMatterCompression],
  "dark-matter decay ratio : " <> nf[darkMatterDecayRatio],
  "early dark-energy tail : " <> nf[earlyDarkEnergyTail],
  "late dark-energy tail : " <> nf[lateDarkEnergyTail],
  "dark-energy tail ratio : " <> nf[darkEnergyTailRatio],
  "early Hubble proxy : " <> nf[earlyHubbleProxy],
  "late Hubble proxy : " <> nf[lateHubbleProxy],
  "Hubble proxy ratio : " <> nf[hubbleProxyRatio],
  "dark-sector pass : " <> yn[darkSectorPass],
  "",
  "MODULE C : BBN PRIMORDIAL PROXY",
  "initial Yp : " <> nf[initialYp],
  "final Yp : " <> nf[finalYp],
  "delta Yp : " <> nf[deltaYp],
  "gamma-zeta initial : " <> sci[gammaZetaInitial],
  "gamma-zeta final : " <> sci[gammaZetaFinal],
  "BBN proxy pass : " <> yn[bbnProxyPass],
  "",
  "MODULE B : CP / EDM PROXY",
  "theta max : " <> sci[thetaMax],
  "theta mean abs : " <> sci[thetaMeanAbs],
  "CP residual : " <> sci[cpResidual],
  "EDM suppression N18 : " <> sci[fractalEDMSuppressionN18],
  "EDM suppression N40 : " <> sci[fractalEDMSuppressionN40],
  "CP/EDM pass : " <> yn[cpEDMPass],
  ""
}

```

```
"MODULE F/J : OBSERVER BRANCHING",
"observer divergence : " <> sci[observerDivergence],
"branch separation max : " <> sci[branchSeparationMax],
"branch separation final : " <> sci[branchSeparationFinal],
"decoherence proxy : " <> sci[decoherenceProxy],
"observer pass : " <> yn[observerBranchingPass],
"",
"MODULE K : COLLAPSE-REBIRTH MEMORY",
"rebirth initial : " <> nf[rebirthInitial],
"rebirth final : " <> nf[rebirthFinal],
"rebirth boundary gap : " <> nf[rebirthBoundaryGap],
"best lag : " <> nf[bestLag],
"best lag correlation : " <> nf[bestLagCorrelation],
"memory transfer score : " <> nf[memoryTransferScore],
"collapse-rebirth pass : " <> yn[collapseRebirthPass],
"",
"MODULE E/H : FRACTAL GEOMETRY",
"normalized psi RMS : " <> nf[normalizedPsiRMS],
"fractal coherence proxy : " <> nf[fractalCoherenceProxy],
"spin-foam initial weight : " <> nf[spinFoilWeightInitial],
"spin-foam final weight : " <> nf[spinFoilWeightFinal],
"geometry coherence mean : " <> nf[geometryCoherenceMean],
"geometry pass : " <> yn[geometryPass],
"",
"CLOSURE",
"combined projection pass : " <> yn[combinedProjectionPass],
"",
"INTERPRETATION",
"The combined downstream physical-projection screen propagates the",
"frozen G/R/N packet through RFC dark-sector, primordial-abundance,",
"CP/asymmetry, observer-branching, collapse-rebirth, and geometry",
"proxies without retuning. Outputs remain internal/projected RFC",
"quantities, not final external laboratory validation."
},
"\n";

Framed[
Style[out, FontFamily -> "Courier", 14],
Background -> Lighter[Gray, .96],
FrameStyle -> GrayLevel[.75],
RoundingRadius -> 3,
ImageSize -> 330
]
```

J.17 Dimensionless observable validation layer

This lane is carried as a closed ledger in the canonical assembly packet. The raw standalone Wolfram packet is not separately expanded in the preserved closeout master, so the carried status is reproduced from TK2.

```
closeout pass count : 17/17
all closeouts pass : YES
```

J.18 TK2 - Canonical preprint assembly ledger

```
protocol : TK2
basis : Completed RFC closeout ledgers + triad-first preprint scaffold
target : canonical preprint assembly ledger
title : Recursive Fractal Cosmology: A Generative Ontology of Existence
author : Allan Edward
score : 1.000
final : FINAL-TK2-CANONICAL-PREPRINT-ASSEMBLY-PASS / MANUSCRIPT-ASSEMBLY-BOUNDARY
```

```
FROZEN RFC INPUT PACKET
delta : 4.669
cycleLength : 60.000
triad : 3.000
nClosure : 18.000
nFullCanon : 40.000
alpha : 0.026
nu : 0.42e-2
epsilon : 0.11e-3
lambdaNorm : 0.489
targets used : NO
```

CLOSEOUT LEDGER

1. Module G : Deterministic Triadic Closure : YES
2. Module R V3 : Source-Coupled Triad Closure Audit : YES
3. Module N V2 : Dimensional Projection Bridge : YES
4. Module S/T : SI Bridge and Dimensionless Coupling Map : YES
5. QG/QG2/J2 : Spin-Foam and Quantum-Geometry Suite : YES
6. Y3 : Frozen Particle-Sector Retest and CKM theta13 Repair : YES
7. U/U2-U5 : Standard-Model-Facing Closure Ledger : YES
8. P2 : Constants / Parameter-Table Boundary : YES
9. V2 : Precision Cosmology Boundary : YES
10. E2 : Early-Universe Relic / Transition Boundary : YES
11. W2 : BBN / Li7 Boundary : YES
12. X2 : CP / EDM / Baryogenesis Boundary : YES
13. GW2 : Gravitational-Wave / Ringdown Boundary : YES
14. Z2 : Observer / Neural / EEG Boundary : YES
15. LEG2 : Legacy Reconciliation Ledger : YES

16. Combined Downstream Physical-Projection Screen : Internal projected-domain screen : YES
17. Dimensionless Observable Validation Layer : Internal dimensionless validation layer : YES
closeout pass count : 17/17
all closeouts pass : YES

CANONICAL PREPRINT SECTION ORDER

1. Abstract / Claim-Boundary Summary : YES
2. Motivation for the Canonical Rebuild : YES
3. CIF / QV / RFL Triad Foundation : YES
4. First Triadic Action: QV(CIF) -> RFL : YES
5. Recursive Kernel and Modal Basis : YES
6. Field / Module Provenance : YES
7. Module G - Deterministic Triadic Closure : YES
8. Module R V3 - Source-Coupled Triad Closure Audit : YES
9. Module N V2 - Dimensional Projection Bridge : YES
10. Combined Downstream Physical-Projection Screen : YES
11. Dimensionless Observable Validation Layer : YES
12. Module S/T - SI Bridge and Dimensionless Coupling Map : YES
13. QG/QG2/J2 - Spin-Foam and Quantum-Geometry Suite : YES
14. Y3 - Frozen Particle-Sector Retest and CKM theta13 Repair : YES
15. U/U2-U5 - Standard-Model-Facing Closure Ledger : YES
16. P2 - Constants / Parameter-Table Boundary : YES
17. V2 - Precision Cosmology Boundary : YES
18. E2 - Early-Universe Relic / Transition Boundary : YES
19. W2 - BBN / Li7 Boundary : YES
20. X2 - CP / EDM / Baryogenesis Boundary : YES
21. GW2 - Gravitational-Wave / Ringdown Boundary : YES
22. Z2 - Observer / Neural / EEG Boundary : YES
23. LEG2 - Legacy Reconciliation Ledger : YES
24. Open-Ledger Watchlist : YES
25. Global Falsifiers / Blocked Claims / External-Run Registry : YES
26. Source Manifest / Reproducibility Checklist : YES
27. References and Reproducibility Strategy : YES
28. Appendix: Closeout Ledgers and Wolfram Reproducibility Packets : YES
section count : 28
section order pass : YES

SOURCE MANIFEST

source 1 : 4 canonical preprint PDFs : authoritative RFC theory source : YES
source 2 : The rebuild.docx : new preprint scaffold and deterministic rebuild logic : YES
source 3 : Simulation Meta Data 1 : QG/QG2/J2 finite spin-foam theorem-support ladder : YES
source 4 : Simulation Meta Data 2 : A-Z closeout and external-boundary ledgers : YES
source 5 : Simulation pictures : visual notebook output trace : YES
source 6 : Module G/R/N/S/T frozen packet : current active deterministic spine : YES
source 7 : SimulationConfigs JSON : machine-readable module catalog : YES
source 8 : Validation screens JSON : downstream validation-screen summaries : YES
source 9 : PDE kernels : kernel and operator documentation : YES
source 10 : Claim boundaries : blocked claims and external-run boundaries : YES
source manifest pass : YES

ASSEMBLY PRINCIPLES

triad first : YES
old MCMC deprecated : YES
modules as downstream projections : YES
frozen packet first : YES
no-retune rule preserved : YES
claim boundaries preserved : YES
external validation separated : YES
Wolfram packets appendix ready : YES
references strategy ready : YES
reproducibility checklist ready : YES
assembly principles pass : YES

MANUSCRIPT STATUS BOUNDARY

manuscript assembly ready : YES
appendix assembly ready : YES
source-backed : YES
preprint draft complete : NO
journal submission ready : NO
peer review complete : NO
independent replication complete : NO
external validation complete : NO
full laboratory theory claimed : NO
boundary safe : YES

FALSIFIERS

claim draft already complete fails : YES
claim journal ready fails : YES
claim peer review complete fails : YES
claim independent replication complete fails: YES
claim external validation complete fails : YES
claim full lab theory fails : YES
claim old MCMC active fails : YES
claim retune allowed fails : YES
claim no boundary needed fails : YES
falsifier pass : YES

CLOSURE

ledger pass : YES

INTERPRETATION

TK2 closes the canonical preprint assembly ledger.
It confirms that the module closeouts, source manifest, preprint section order, Wolfram reproducibility appendix, claim-boundary system, and reference/reproducibility strategy are ready to assemble into the new RFC preprint.

PASS means manuscript-assembly-ready and appendix-assembly-ready.
It does not mean the final preprint draft is already complete, journal submission ready, peer reviewed, independently replicated, externally validated, or a completed laboratory theory.

1. Module G - Deterministic Triadic Closure

```
2. Module R V3 - Source-Coupled Triad Closure Audit
3. Module N V2 - Dimensional Projection Bridge
4. Module S/T - SI Bridge and Dimensionless Coupling Map
5. QG/QG2/J2 - Spin-Foam and Quantum-Geometry Suite
6. Y3 - Frozen Particle-Sector Retest and CKM theta13 Repair
7. U/U2-U5 - Standard-Model-Facing Closure Ledger
8. P2 - Constants / Parameter-Table Boundary
9. V2 - Precision Cosmology Boundary
10. E2 - Early-Universe Relic / Transition Boundary
11. W2 - BBN / Li7 Boundary
12. X2 - CP / EDM / Baryogenesis Boundary
13. GW2 - Gravitational-Wave / Ringdown Boundary
14. Z2 - Observer / Neural / EEG Boundary
15. LEG2 - Legacy Reconciliation Ledger
16. Combined Downstream Physical-Projection Screen
17. Dimensionless Observable Validation Layer
18. TK2 - Canonical Preprint Assembly Ledger
'''
```


References

- [1] M. J. Feigenbaum, Quantitative universality for a class of nonlinear transformations, *Journal of Statistical Physics* 19, 25-52 (1978).
- [2] M. J. Feigenbaum, The transition to aperiodic behavior in turbulent systems, *Communications in Mathematical Physics* 77, 65-86 (1980).
- [3] R. M. May, Simple mathematical models with very complicated dynamics, *Nature* 261, 459-467 (1976).
- [4] B. B. Mandelbrot, *The Fractal Geometry of Nature*. W. H. Freeman (1982).
- [5] Planck Collaboration, Planck 2018 results. VI. Cosmological parameters, *Astronomy and Astrophysics* 641, A6 (2020), arXiv:1807.06209.
- [6] DESI Collaboration, DESI 2024 VI: Cosmological constraints from the measurements of baryon acoustic oscillations, arXiv:2404.03002 (2024).
- [7] D. Brout et al., The Pantheon+ Analysis: Cosmological constraints, *Astrophysical Journal* 938, 110 (2022), arXiv:2202.04077.
- [8] D. Blas, J. Lesgourgues, and T. Tram, The Cosmic Linear Anisotropy Solving System (CLASS). Part II: approximation schemes, *JCAP* 07, 034 (2011), arXiv:1104.2933.
- [9] A. Lewis, A. Challinor, and A. Lasenby, Efficient computation of CMB anisotropies in closed FRW models, *Astrophysical Journal* 538, 473 (2000), arXiv:astro-ph/9911177.
- [10] J. Torrado and A. Lewis, Cobaya: code for Bayesian analysis of hierarchical physical models, *JCAP* 05, 057 (2021), arXiv:2005.05290.
- [11] B. Audren, J. Lesgourgues, K. Benabed, and S. Prunet, Conservative constraints on early cosmology with Monte Python, *JCAP* 02, 001 (2013), arXiv:1210.7183.
- [12] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, CODATA recommended values of the fundamental physical constants: 2022, arXiv:2409.03787 (2024); NIST constants database.
- [13] Bureau International des Poids et Mesures, *The International System of Units (SI Brochure)*, 9th edition (2019, updated online).
- [14] Particle Data Group, Review of Particle Physics, *Progress of Theoretical and Experimental Physics* 2024, 083C01 (2024).
- [15] I. Esteban et al., NuFIT 6.0 global analysis of neutrino oscillation data (2024), www.nu-fit.org.
- [16] A. D. Sakharov, Violation of CP invariance, C asymmetry, and baryon asymmetry of the universe, *JETP Letters* 5, 24-27 (1967).
- [17] M. Kobayashi and T. Maskawa, CP-violation in the renormalizable theory of weak interaction, *Progress of Theoretical Physics* 49, 652-657 (1973).
- [18] Z. Maki, M. Nakagawa, and S. Sakata, Remarks on the unified model of elementary particles, *Progress of Theoretical Physics* 28, 870-880 (1962).
- [19] ACME Collaboration, Improved limit on the electric dipole moment of the electron, *Nature* 562, 355-360 (2018).
- [20] C. Abel et al., Measurement of the permanent electric dipole moment of the neutron, *Physical Review Letters* 124, 081803 (2020).

- [21] A. H. Guth, Inflationary universe: a possible solution to the horizon and flatness problems, *Physical Review D* 23, 347-356 (1981).
- [22] A. D. Linde, A new inflationary universe scenario, *Physics Letters B* 108, 389-393 (1982).
- [23] A. A. Starobinsky, A new type of isotropic cosmological models without singularity, *Physics Letters B* 91, 99-102 (1980).
- [24] E. W. Kolb and M. S. Turner, *The Early Universe*. Addison-Wesley (1990).
- [25] A. Vilenkin and E. P. S. Shellard, *Cosmic Strings and Other Topological Defects*. Cambridge University Press (1994).
- [26] B. J. Carr and S. W. Hawking, Black holes in the early Universe, *Monthly Notices of the Royal Astronomical Society* 168, 399-415 (1974).
- [27] S. Seager, D. D. Sasselov, and D. Scott, A new calculation of the recombination epoch, *Astrophysical Journal* 523, L1-L5 (1999).
- [28] J. Chluba and R. M. Thomas, Towards a complete treatment of the cosmological recombination problem, *Monthly Notices of the Royal Astronomical Society* 412, 748-764 (2011).
- [29] O. Pisanti et al., PArthENoPE: Public Algorithm Evaluating the Nucleosynthesis of Primordial Elements, *Computer Physics Communications* 178, 956-971 (2008).
- [30] C. Pitrou, A. Coc, J.-P. Uzan, and E. Vangioni, Precision big bang nucleosynthesis with improved helium-4 predictions, *Physics Reports* 754, 1-66 (2018).
- [31] C. Rovelli, *Quantum Gravity*. Cambridge University Press (2004).
- [32] A. Ashtekar and J. Lewandowski, Background independent quantum gravity: a status report, *Classical and Quantum Gravity* 21, R53 (2004).
- [33] J. Engle, R. Pereira, and C. Rovelli, The loop-quantum-gravity vertex amplitude, *Physical Review Letters* 99, 161301 (2007).
- [34] L. Freidel and K. Krasnov, A new spin foam model for 4D gravity, *Classical and Quantum Gravity* 25, 125018 (2008).
- [35] A. Connes, *Noncommutative Geometry*. Academic Press (1994).
- [36] H. Everett, Relative state formulation of quantum mechanics, *Reviews of Modern Physics* 29, 454-462 (1957).
- [37] W. H. Zurek, Decoherence, einselection, and the quantum origins of the classical, *Reviews of Modern Physics* 75, 715-775 (2003).
- [38] LIGO Scientific Collaboration and Virgo Collaboration, Observation of gravitational waves from a binary black hole merger, *Physical Review Letters* 116, 061102 (2016).
- [39] R. Abbott et al., Open data from the first and second observing runs of Advanced LIGO and Advanced Virgo, *SoftwareX* 13, 100658 (2021).
- [40] A. Gramfort et al., MEG and EEG data analysis with MNE-Python, *Frontiers in Neuroscience* 7, 267 (2013).
- [41] J. M. Beggs and D. Plenz, Neuronal avalanches in neocortical circuits, *Journal of Neuroscience* 23, 11167-11177 (2003).
- [42] A. Clauset, C. R. Shalizi, and M. E. J. Newman, Power-law distributions in empirical data, *SIAM Review* 51, 661-703 (2009).